**INSTRUCTIONS**

# AF3060 INVERTER DRIVES

DRIVER - 331X358BAG01, G02

## OPERATION – TROUBLE SHOOTING – REPAIR

GEK-62512A

REV. 1 6/21/77

| | | |
|---|---|---|
| DRIVER ASSEMBLY | ELEM | 36D865533BA |
| | CONN | 36C764839BA |

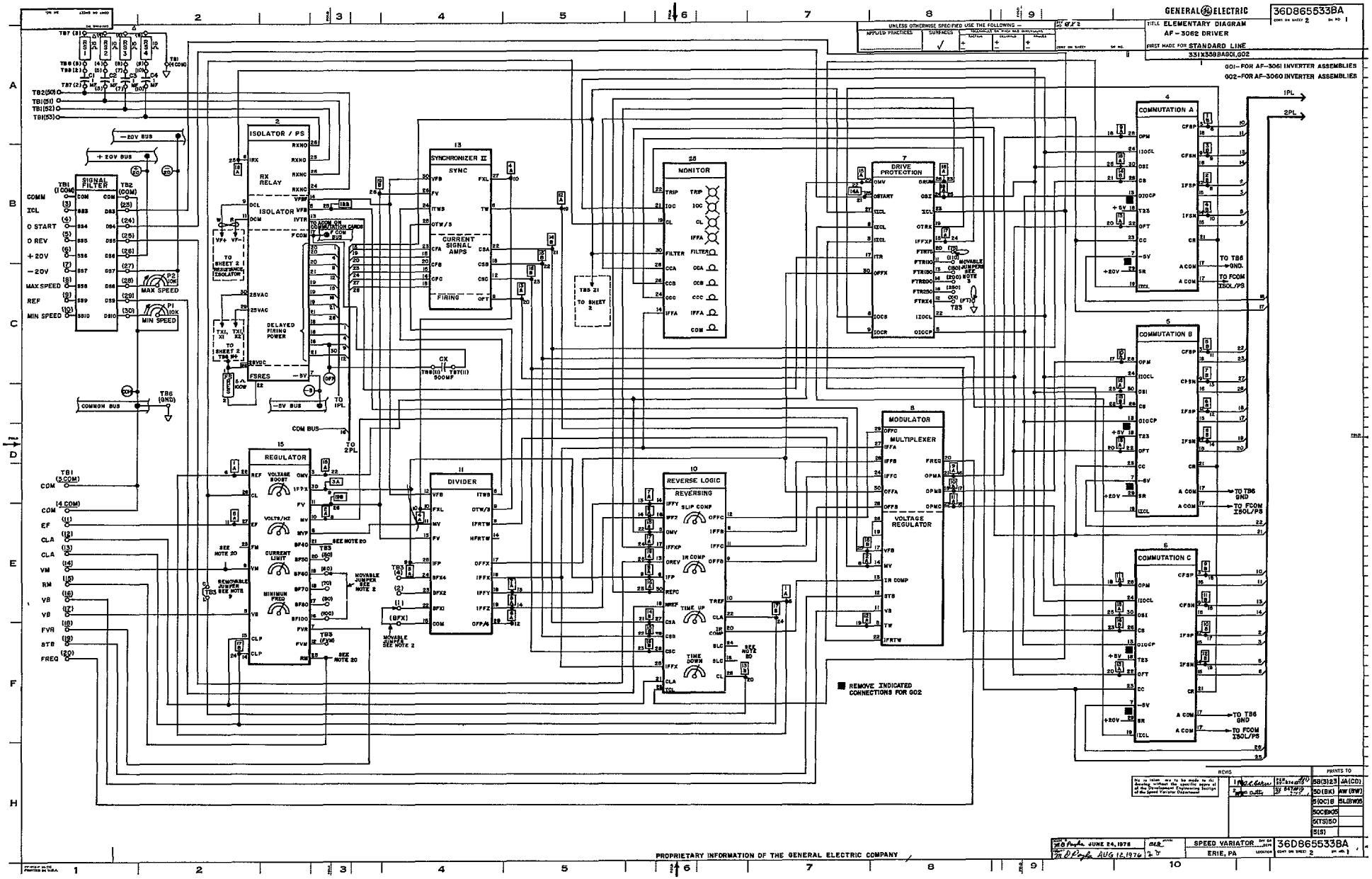
PRINTED CIRCUIT CARD DIAGRAMS

| | |
|---|---|
| DRIVE PROTECTION | 36C762849AC |
| COMMUTATION | 36C764126AC |
| REGULATOR | 36C764136AB |
| MODULATOR | 36C764107AA |
| SYNCHRONIZER II | 36C764108AA |
| REVERSE LOGIC | 36C762855AB |
| SIGNAL FILTER | 36C762868AA |
| MONITOR | 36C762854AA |
| ISOLATOR/POWER SUPPLY | 36C762869CA |
| 5V POWER SUPPLY | 36C762870AB |
| 20V POWER SUPPLY | 36C762896AB |
| TEST CARD | 36C762858AA |
| DIVIDER | 36C762895AB |
| POWER INTERFACE | 36C764129AA |
| RESISTANCE ISOLATOR | 36C764140AA |

*These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the General Electric Company.*

GENERAL ELECTRIC

ademark of the General Electric Co.


GENERAL ELECTRIC
36D865533BA
TITLE ELEMENTARY DIAGRAM
AF-3062 DRIVER
FIRST MADE FOR STANDARD LINE
331X358BA001,002
001-FOR AF-3061 INVERTER ASSEMBLIES
002-FOR AF-3060 INVERTER ASSEMBLIES
SIGNAL FILTER
ISOLATOR / PS
RX RELAY
ISOLATOR
SYNCHRONIZER II
SYNC
CURRENT SIGNAL AMPS
FIRING
DELAYED FIRING POWER
MONITOR
DRIVE PROTECTION
COMMUTATION A
COMMUTATION B
COMMUTATION C
REGULATOR
DIVIDER
REVERSE LOGIC
REVERSING
SLIP COMP
IR COMP
TIME UP
TIME DOWN
MODULATOR
MULTIPLEXER
VOLTAGE REGULATOR
REMOVE INDICATED CONNECTIONS FOR 002
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY
SPEED VARIATOR
ERIE, PA
36D865533BA

G01–FOR AF-3061 INVERTER ASSEMBLIES
G02–FOR AF-3060 INVERTER ASSEMBLIES

## DRIVER NOMENCLATURE

### NOMENCLATURE

THE ABBREVIATED NOMENCLATURE USED ON THE DRIVER IS LISTED BELOW. A "1" (+5 VOLTS) OR A "0" (ZERO VOLTS) PRECEDING THE ABBREVIATION INDICATES A LOGIC SIGNAL AND REFERS TO THE MEANINGFUL LOGIC STATE OR THE LOGIC STATE WHEN THE CORRESPONDING PHASE IN THE POWER INVERTER IS POSITIVE. PHASES A, B, AND C REFER TO THE POWER PHASES. PHASES X, Y, AND Z REFER TO THE DRIVER PHASES BEFORE THEY ARE COMMITTED TO A POWER PHASE. THE FOLLOWING RELATION EXISTS BETWEEN THE DRIVER AND POWER PHASES.

| DRIVER PHASE | POWER PHASE | |
|---|---|---|
| | FORWARD | REVERSE |
| X | A | A |
| Y | B | C |
| Z | C | B |

EXAMPLES:

- 110CL – INSTANTANEOUS OVERCURRENT LOCKOUT, LOCKS OUT FIRING SIGNALS WHEN THE "1" STATE.
- 0FFX – THE FUNDAMENTAL DRIVER FREQUENCY OF PHASE X (POWER PHASE A). THIS SIGNAL IS IN THE "0" STATE WHEN PHASE A IS POSITIVE.

| | |
|---|---|
| 8FX1 | BASE FREQUENCY TIMES 1 |
| 8FX2 | BASE FREQUENCY TIMES 2 |
| 8FX4 | BASE FREQUENCY TIMES 4 |
| 8FN(50,60,ETC.) | BASE FREQUENCY = 40 HZ (50 HZ, 60 HZ, ETC.) |
| CCA(B,C) | COMMUTATION CURRENT - PHASE A (B,C) |
| CFA (B,C) | CURRENT FEEDBACK - PHASE A (B,C) |
| CFSN | COMMUTATION FIRING SIGNAL - NEGATIVE BUS |
| CFSP | COMMUTATION FIRING SIGNAL - POSITIVE BUS |
| CL | CURRENT LIMIT |
| CLA | CURRENT LIMIT ADJUST |
| CLP | CURRENT LIMIT POTENTIOMETER |
| CR | COMMUTATION RECALIBRATION |
| CSA (B,C) | CURRENT SIGNAL - PHASE A (B,C) |
| DCL | D.C. LINK |
| DCM | D.C. MIDPOINT |
| DFP | DELAYED FIRING POWER |
| EF | EXTERNAL FREQUENCY |
| 0FFA | FUNDAMENTAL FREQUENCY - PHASE A |
| 1FFA | FUNDAMENTAL FREQUENCY - PHASE A |
| 0FFB | FUNDAMENTAL FREQUENCY - PHASE B |
| 1FFB | FUNDAMENTAL FREQUENCY - PHASE B |
| 0FFC | FUNDAMENTAL FREQUENCY - PHASE C |
| 1FFC | FUNDAMENTAL FREQUENCY - PHASE C |
| FILTER | FILTER CURRENT FEEDBACK |
| 0FFX | FUNDAMENTAL FREQUENCY - PHASE X |
| 1FFX | FUNDAMENTAL FREQUENCY - PHASE X |
| 1FFXP | A "1" PULSE RELATED TO PHASE X |
| 1FFY | FUNDAMENTAL FREQUENCY - PHASE Y |
| 1FFZ | FUNDAMENTAL FREQUENCY - PHASE Z |
| 1FP | FREQUENCY PULSE |
| 0FP/2 | FREQUENCY PULSE DIVIDED BY 2 |
| FM | FREQUENCY MODIFICATION |
| FREQ | FREQUENCY PULSE READOUT |
| 1FRTW | FREE RUNNING TRIANGLE WAVE |
| OFT | FIRING TRAIN |
| FTR75(110,150,ETC) | FREQUENCY TRIP = 75 HZ. (110 HZ, 150 HZ, ETC) |
| FTRX4 | FREQUENCY TRIP - TIMES FOUR |
| FV | FREQUENCY VOLTAGE |
| FVR | FREQUENCY VOLTAGE READOUT |
| FXL | FREQUENCY TIMES DC LINK |
| FVM | FREQUENCY VOLTAGE MODIFICATION |
| 1CL | INSTANTANEOUS CURRENT LIMIT |
| 1FSN | INVERTER FIRING SIGNAL - NEGATIVE BUS |
| 1FSP | INVERTER FIRING SIGNAL - POSITIVE BUS |
| HFTRW | HIGH FREQUENCY TRIANGLE WAVE |
| 110CL | INSTANTANEOUS OVERCURRENT LOCKOUT |
| RM | REFERENCE MODIFICATION |
| 01OCP | INSTANTANEOUS OVERCURRENT PULSE |
| 10CR | INSTANTANEOUS OVERCURRENT RECALIBRATION |
| 10CS | INSTANTANEOUS OVERCURRENT SIGNAL |
| I.R. COMP. | MOTOR RESISTANCE COMPENSATION |
| OMV | MINIMUM VOLTAGE |
| MV | MODULATOR VOLTAGE |
| NREF | REFERENCE |
| OPMA (B,C) | PULSE MODULATION SIGNAL - PHASE A (B,C) |
| REF | REFERENCE |
| REFC | REFERENCE CLAMP |
| OREV | REVERSE |
| ORUN | RUN SIGNAL RECEIVED |
| 1RX | RX RELAY - PICKS UP ON "1" SIGNAL |
| RXNC | RX RELAY - NORMALLY CLOSED CONTACTS |
| RXNO | RX RELAY - NORMALLY OPEN CONTACTS |
| OSI | STOP INVERTER |
| 1TWS | TRIANGLE WAVE SYNCH |
| OTW/3 | TRIANGLE WAVE DIVIDED BY 3 |
| T23 | 23 MICROSECOND TIMER |
| SLC | SLIP COMPENSATION |
| OSTART | START |
| 1TR | DRIVE TRIP ON "1" PULSE |
| TREF | TIMED REFERENCE |
| OTRX | TRIP EXECUTE |
| TW | TRIANGLE WAVE |
| VB | VOLTAGE BOOST |
| VFB | VOLTAGE FEEDBACK |
| 1VTR | VOLTAGE TRIP |
| SR | STEEP RAMP |
| STB | STABILITY |
| VM | VOLTAGE MODIFICATION |

## DRIVER NOTES

1. THE NUMBERS ABOVE THE CARD BLOCKS REFER TO THE LOCATION OF THE CARD IN THE DRIVER RACK.
2. BASE FREQUENCY IS THE FREQUENCY AT WHICH THE DRIVE REACHES FULL VOLTAGE. THE BASE FREQUENCY IS SELECTED BY A BASE FREQUENCY JUMPER CONNECTED TO TB3 (100) AND A BASE FREQUENCY MULTIPLIER JUMPER CONNECTED TO TB3 (0FX). THE JUMPERS SHOWN ARE FOR A 30 HZ DRIVE. SEE THE TABLE BELOW FOR PROPER PLACEMENT OF THESE JUMPERS. CHECK DRIVE ELEMENTARY DIAGRAM FOR PROPER PLACEMENT OF THIS JUMPER. CAUTION: IMPROPER PLACEMENT OF THIS JUMPER MAY RESULT IN DRIVE MALFUNCTION OR DAMAGE. DO NOT MAKE ANY CHANGES WITHOUT CONSULTING GENERAL ELECTRIC COMPANY.

| TB3 JUMPERS 100 / 0FX | HERTZ (40) | (50) | (60) | (70) | (80) | (100) |
|---|---|---|---|---|---|---|
| (1) | 20 | 25 | 30 | 35 | 40 | 50 |
| (2) | 40 | 50 | 60 | 70 | 80 | 100 |
| (4) | 80 | 100 | 120 | 140 | 160 | |

3. JUMPERS ANCHORED TO TB3(75) AND (X4) SELECT THE FREQUENCY TRIP AND ARE CONNECTED AS SHOWN IN THE TABLE BELOW. CHECK THE DRIVE ELEMENTARY DIAGRAM FOR PROPER PLACEMENT OF THESE JUMPERS. WARNING: IMPROPER PLACEMENT OF THESE JUMPERS MAY RESULT IN DRIVE MALFUNCTION OR PRESENT AN EQUIPMENT OR PERSONNEL HAZARD DUE TO MOTOR OVERSPEED. DO NOT MAKE ANY CHANGES WITHOUT CONSULTING GENERAL ELECTRIC COMPANY.

| TB3(75) JUMPER PLACEMENT | 75 | 110 | 150 | 200 | 250 |
|---|---|---|---|---|---|
| FREQUENCY TRIP (HZ) | 75 | 110 | 150 | 200 | 200 |
| WITH TB3(FT) JUMPERED TO (X4) | 300 | 440 | 600 | 800 | 1000 |

4. SEVERAL DRIVER CIRCUITS ARE WIRED TO THE MONITOR CARD TEST CONNECTORS TO ENABLE EXTERNAL CONNECTIONS OF TEST PROBES.

   WHEN LIT, THE LIGHTS ON THE MONITOR CARD MAKE THE FOLLOWING INDICATIONS.

| LIGHT | INDICATION |
|---|---|
| TRIP | DRIVE TRIP DUE TO UNDERVOLTAGE, OVERVOLTAGE OR HIGH FREQUENCY (IOC WILL ALSO LIGHT) |
| IOC | SHOOT THROUGH OR OTHER MAJOR POWER SYSTEM FAULT |
| CL | DRIVE IS IN CURRENT LIMIT |
| 1FFA | PHASE A FREQUENCY SIGNAL |

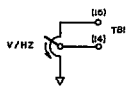
5. TEST RECEPTACLES "A" AND "B" SHARE ONE TEST CARD WHICH HAS A HOME LOCATION AT "A", POSITION NO.23. THESE RECEPTACLES AND TEST CARD ARE USED TO MONITOR THE DRIVE CIRCUITS WIRED TO THEM.

   A WIRE PERMANENTLY CONNECTED TO TAB 28 OF BOTH TEST RECEPTACLES HAS A MOVEABLE TEST CONNECTOR ON THE OTHER END WHICH MAY BE CONNECTED TO ANY POINT IN THE BACK PLANE WIRING. THIS WIRE IS CONNECTED TO THE "SEL" TEST CONNECTOR FOR MONITORING ANY POINT DESIRED.

   SINCE MANY OF THE TEST POINTS ARE LOGIC SIGNALS, PROPERLY SHIELDED TEST LEADS SHOULD BE USED AT ALL TIMES.

6. THE "MIN. FREQ." POTENTIOMETER P268 ON THE REGULATOR CARD ADJUSTS THE MINIMUM FREQUENCY. A REFERENCE SETTING BELOW THE MINIMUM FREQUENCY WILL REDUCE THE OUTPUT VOLTAGE BUT NOT THE FREQUENCY. THIS WILL PROVIDE SMOOTH STARTUP AND REVERSING.
7. THE "VOLTS/HERTZ VERNIER" POTENTIOMETER P267 ON THE REGULATOR CARD ADJUSTS THE VOLTS PER HERTZ OF THE INVERTER WITHIN ±10% OF NOMINAL.

   AN EXTERNAL VOLTS PER HERTZ POTENTIOMETER MAY BE ADDED AS INDICATED BELOW. THIS POTENTIOMETER WILL ALLOW AN EXTERNAL VOLTS/HZ ADJUSTMENT OF 2:1 AS AN INCREASE OVER THE INTERNAL SETTING.




8. IF A VOLTAGE BOOST FUNCTION IS PROVIDED THE "VOLTAGE BOOST" POTENTIOMETER P265 ON THE REGULATOR CARD CAN BE USED TO ADD A FIXED VOLTAGE TO THE INVERTER INDEPENDENT OF FREQUENCY. AN EXTERNAL VOLTAGE BOOST POTENTIOMETER MAY BE PROVIDED.
9. THE "CURRENT LIMIT" POTENTIOMETER P266 ON THE REGULATOR CARD ADJUSTS THE CURRENT LIMIT BETWEEN 50% AND 150% OF RATED DRIVE CURRENT. AN ADDITIONAL EXTERNAL CURRENT LIMIT RHEOSTAT MAY BE PROVIDED TO DECREASE THE CURRENT LIMIT SET BY P266. CURRENT LIMIT MAY BE RENDERED INOPERATIVE BY REMOVING THE "CL" JUMPER ON TB3.
10. THE "I.R. COMP." POTENTIOMETER P476 ON THE REVERSE LOGIC CARD MAY BE ADJUSTED TO COMPENSATE FOR VOLTAGE LOSS DUE TO MOTOR RESISTANCE. IF NOT USED, TURN FULL CCW.
11. IF SLIP COMPENSATION IS PROVIDED THE "SLIP COMP." POTENTIOMETER P475 ON THE REVERSE LOGIC CARD MAY BE ADJUSTED TO COMPENSATE FOR SLIP IN MOTOR SPEED DUE TO LOADING.
12. FOR A VOLTAGE REFERENCE THE "TIME UP", P477 AND "TIME DOWN", P478 POTENTIOMETERS ON THE REVERSE LOGIC CARD ADJUST THE ACCELERATION AND DECELERATION RATES RESPECTIVELY. THE TIME FOR THE DRIVE TO RESPOND TO A 10V STEP CHANGE IN REFERENCE MAY BE VARIED BETWEEN 2.0 SEC. AND 32 SEC.
13. THE FAST CURRENT LIMIT SETTING IS FIXED AND SET AT A PREDETERMINED VALUE APPROXIMATELY 200% OF PEAK RATED CURRENT. THIS IS A "KEEP RUNNING" LIMIT THAT REDUCES VOLTAGE BUT DOES NOT TRIP THE DRIVE.
14. CONNECTOR PIN NUMBERS FOR 1PL AND 2PL ARE SHOWN AS FOLLOWS:




THE HEAVY LINES REPRESENT THE CABLES LEADING TO THE FEMALE CONNECTORS 1PL AND 2PL MOUNTED NEAR THE BOTTOM OF THE CARD RACK.

15. 

INDICATES TERMINAL BOARD NUMBER



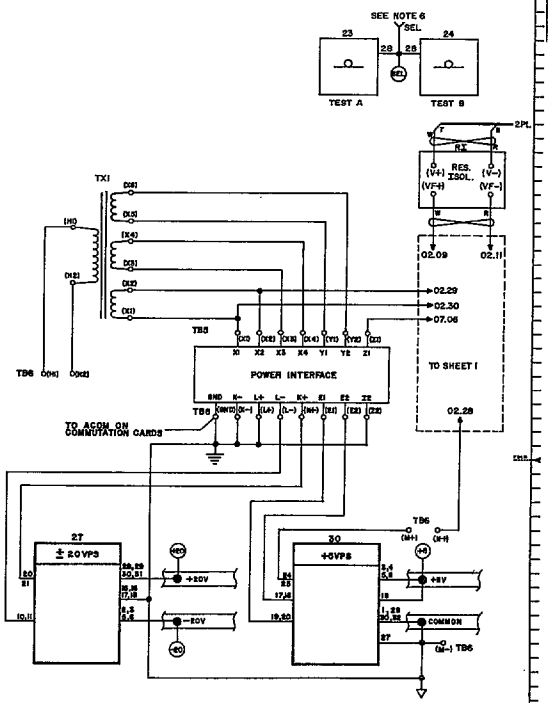
INDICATES TAB NUMBER ON CARD WHICH CORRESPONDS TO MATCHING RECEPTACLE NUMBER.

16. POTENTIOMETER ARROWS INDICATE THE WIPER DIRECTION ON THE ELEMENTARY DIAGRAM AS THE POTENTIOMETER SHAFT IS ROTATED CLOCKWISE.
17.  INDICATES TEST CONNECTOR ON TEST CARD.
18.  NUMBER IN SYMBOL CORRESPONDS TO INDICATOR POSITION ON THE TEST CARD. LETTER IN SYMBOL CORRESPONDS TO POSITION TEST CARD IS LOCATED. NUMBER BELOW THE SYMBOL INDICATES THE TAB NUMBER IN THE TEST CARD.
19.  SYMBOL FOR TEST CONNECTOR.
20. THE FOLLOWING MODIFICATIONS MAY BE MADE USING RED NYLON JACKETED #24 WIRE:

    15.09 - 15.25 DOUBLES REFERENCE TO VOLTAGE GAIN SUCH AS FOR WIDE CONSTANT HORSE POWER RANGE.

    15.23 - 15.25 DOUBLES REFERENCE TO FREQUENCY GAIN SUCH AS FOR HIGH FREQUENCY DRIVES.

    15.16 - 15.21 FOR 40 HZ BASE FREQUENCY (WITH BASE FREQUENCY JUMPER SELECTION ON 100).



WIRE TABLE

BUS BAR CONNECTIONS

| TAB | CARDS CONNECTED | CIRCUIT |
|---|---|---|
| 02 | 02, 04, 05, 06, 07, 08, 10, 11, 13, 15 | -20V |
| 04 | 02, 04, 05, 06, 07, 08, 10, 11, 13, 15 | +5V |
| | | |
| 07 | 02, 04, 05, 06, 07, | -5V |
| 31 | 02, 04, 05, 06, 07, 08, 10, 11, 13, 15 | +20V |
| 32 | 02, 04, 05, 06, 07, 08, 10, 11, 13, 15 | COM |
| | | |
| 17 | 02, 04, 05, 06 | F COM |

BACK PLANE WIRING

#24 WHITE

02.05-08.17-24.25
02.06-04.30-05.30-06.30-
07.24-23.25
02.13-07.17
02.14-11.12-13.30
02.18-23.30-24.30

04.03-24.06
04.09-24.09
04.12-24.08
04.13-05.13-06.13-07.05
04.14-24.10
04.18-04.04■
04.19-07.27
04.21-05.21-06.21-07.09
04.22-05.22-06.22-13.09-23.20
04.23-25.28
04.24-05.24-06.24-07.22-25.21
04.26-10.27-13.22-24.21
04.28-08.21-23.16
04.29-04.31■

05.03-24.11
05.09-24.13
05.12-24.12
05.14-24.14
05.18-05.04■
05.19-07.03
05.23-25.26
05.26-10.29-13.18-24.22
05.28-08.18-23.17
05.29-05.31■

06.03-24.16
06.09-24.18
06.12-24.17
06.14-24.19
06.18-06.04■
06.19-07.08
06.23-25.24
06.26-10.28-13.12-24.23
06.28-08.23-23.18
06.29-06.31■

07.06-25.30
07.18-10.17-23.24
07.19-25.22
07.23-10.23
07.25-23.21
07.26-10.30-23.23
07.29-10.05-15.03-23.22
07.30-08.30-11.17

08.05-13.08-23.19
08.13-10.20
08.14-15.10-23.08
08.19-08.25
08.22-11.08
08.24-10.11
08.26-10.08
08.27-10.25-11.16-25.14
08.28-10.09
08.29-10.12

10.06-11.25-23.09-15.30
10.10-15.22-23.06
10.13-23.26
10.14-11.18-23.13
10.16-11.19-23.14
10.21-15.13
10.22-15.14-24.24
10.26-24.20

11.06-13.24
11.09-13.25
11.10-13.27-23.10
11.11-15.06
11.13-13.26-15.11-24.26

11.29-23.12

15.26-25.19
15.27-23.11

23.28-24.28

#24 WHITE

+5V

30.03-10.04
30.04-11.04
30.05-13.04
30.05-25.04-24.04-23.04
30.06-15.04
30.06-30.16

-5V

02.07-23.07-24.07-25.07

+20V

27.28-07.31
27.29-08.31
27.30-10.31
27.31-11.31
27.31-24.31-23.31

-20V

27.02-07.02
27.03-08.02
27.05-10.02
27.06-11.02
27.02-24.02-23.02

COMMON

02.17-23.01-24.01-25.01
27.15-05.32
27.16-06.32
27.17-07.32
27.18-08.32

30.01-10.32
30.01-30.27
30.29-11.32
30.30-13.32
30.32-15.32

BACK PANEL WIRING

#14 BLACK

TB1.3COM-TB2.COM-TB6.GND
TB6.GND-TB6.K(MINUS)
TB6.L(PLUS)-TB6.Z2

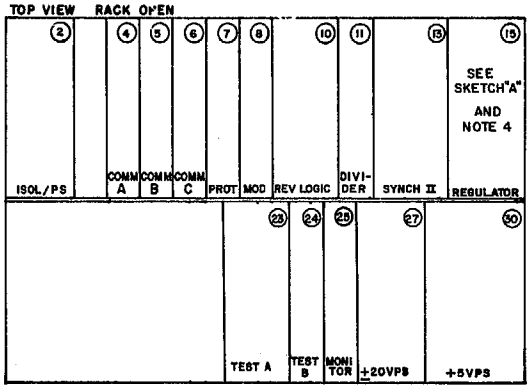
TX1.H1-TB6.H1
TX1.H2-TB6.H2
TX1.X1-TB5.X1
TX1.X2-TB5.X2
TX1.X3-TB5.X3
TX1.X4-TB5.X4
TX1.X5-TB5.Y1
TX1.X6-TB5.Y2

#24 WHITE

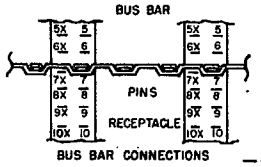
P1.02-P1.03-TB6.GND
P1.01-TB2.30
P2.01-P2.02-TB2.28
P2.03-TB2.27

TB2.50-TB7.1
TB1.51-TB7.4
TB1.52-TB7.6
TB1.53-TB7.9
TB1.4 COM-TB7.2
TB7.3-TB7.7
24.01-TB7.11
24.30-TB8.11

TOP VIEW RACK OPEN




SKETCH "B"



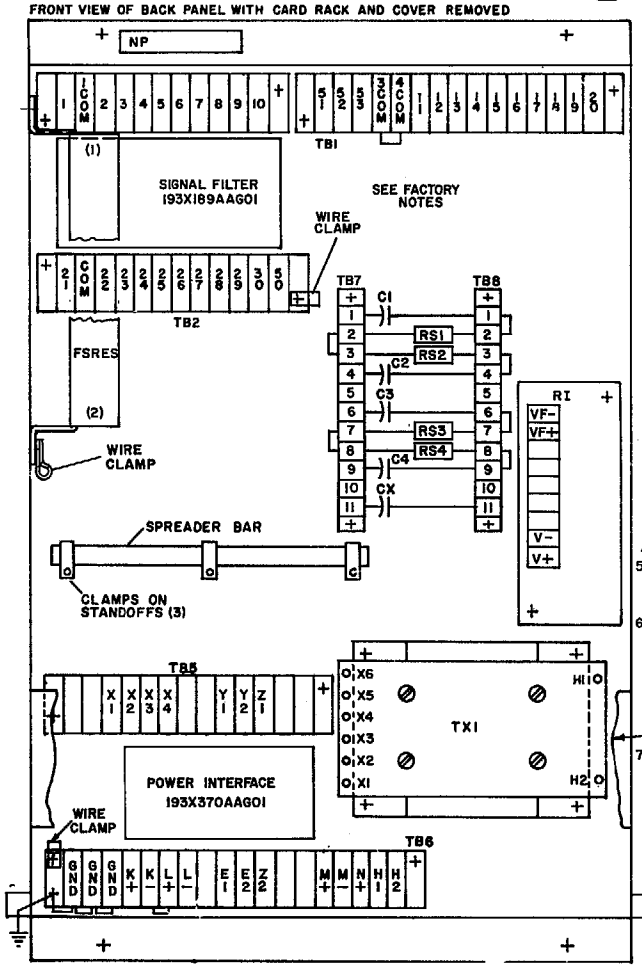

BUS BAR CONNECTIONS

FACTORY NOTES:

1. ASSEMBLE BUS BAR BEFORE WIRING. ALWAYS ASSEMBLE WITH THE BODY OF THE BUS TOWARD THE LOWER NUMBERED SIDE OF THE RECEPTACLE.
2. CONNECT WIRES TO TB1, TB2, & TB3 AT LOWER TERMINALS; ALL WIRES TO TB5 & TB6 AT UPPER TERMINALS.
3. *-CRIMP ALL #24 WIRES ON ONE SPADE TERMINAL.
   **-CRIMP THREE #24 WIRES ON EACH OF TWO SPADE TERMINALS.
4. TB3 HAS BOTH SCREW AND TERMIPOINT CONNECTORS.

TB3 + 50 60 70 80 100 BPX 1 2 4 10 11 + + 75 110 150 200 250 FT X4 FVM C L +

FRONT VIEW OF BACK PANEL WITH CARD RACK AND COVER REMOVED

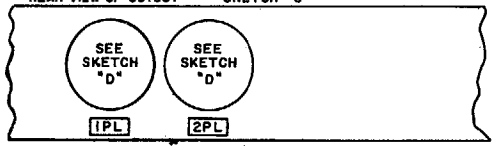


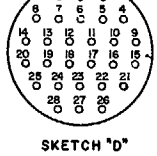

SKETCH "A"
FULL SIZE VIEW OF CARD RECEPTACLE

SEE NOTE 3 ON SH. 2

5. DO NOT BUNDLE WIRES. DISTRIBUTE WIRES ALONG LENGTH OF SPREADER BAR.
6. CUT ALL WIRES APPROXIMATELY ONE INCH LONGER THAN THE SHORTEST POSSIBLE LENGTH.
7. ■ REMOVE INDICATED CONNECTIONS FOR G02.

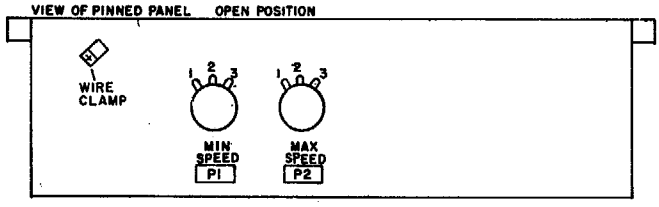
REAR VIEW OF CUTOUT SKETCH "C"



SKETCH "D"

VIEW OF PINNED PANEL OPEN POSITION




PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

DRAWN W.B. Hoyle JUNE 25,1976
APPROVED [illegible] AUG 12,1976
SPEED VARIATOR
ERIE, PA
DIV OR DEPT
LOCATION
36C764839BA
SH. NO. 1 OF 2 SHEETS

REVISIONS

| REV | DESCRIPTION | PER |
|---|---|---|
| 1 | [illegible] | [illegible] |
| 2 | [illegible] | [illegible] |

No revisions are to be made to this drawing without approval of the Development Engineering Section of the Speed Variator Department.

PRINTS TO

| | | | |
|---|---|---|---|
| SST SBO | | JA(CD) | 5A(S) |
| | TB(1) | SL(P4-5) | 5B(BK) |
| | 5GC(S) | AW(BW) | 5B(13,23) |
| | 5GC(BK) | | |

REV NO.

CONT ON SHEET 2

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING—
APPLIED PRACTICES: 7184505
SURFACES
TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± , DECIMALS ± , ANGLES ±

TITLE CONNECTION DIAGRAM
AF3062 DRIVER
FIRST MADE FOR STANDARD LINE 331X358BAG01, G02

G01—FOR AF-3061 INVERTER ASSEMBLIES
G02—FOR AF-3060 INVERTER ASSEMBLIES

GENERAL ELECTRIC
36C764839BA
CONT ON SHEET 2
SH NO 1

## INTERFACE WIRING

### #24 WHITE

TB2.50-02.24
TB1.51-02.25
TB1.52-02.28
TB1.53-02.26

TB1.11-15.27
TB1.12-15.13
TB1.13-10.22
TB1.14-15.09
TB1.15-15.25
TB1.16-08.11
TB1.17-15.05
TB1.18-15.07
TB1.19-08.12
TB1.20-08.20

TB2.23-07.23
TB2.24-07.25
TB2.25-10.13
TB2.26-27.29
TB2.27-27.03
TB2.29-10.19

TB3.50-15.20
TB3.60-15.19
TB3.70-15.18
TB3.80-15.17
TB3.100-15.16
TB3.BFX-11.15

TB3.01-11.22
TB3.02-11.23
TB3.04-11.24
TB3.75-07.20
TB3.110-07.11
TB3.150-07.13
TB3.200-07.14
TB3.250-07.16
TB3.FT-07.15
TB3.X4-07.12
TB3.FVM-15.12
TB3.C-15.26
TB3.L-10.26

TB5.Z1-07.06

TB6.GND-04.17 *
TB6.GND-05.17 *
TB6.GND-06.17 *
TB6.K(PLUS)-27.20 *
TB6.K(PLUS)-27.20 *
TB6.K(PLUS)-27.21 *
TB6.L(MINUS)-27.10 *
TB6.L(MINUS)-27.10 *
TB6.L(MINUS)-27.11 *
TB6.E1-30.19 *
TB6.E1-30.19 *
TB6.E1-30.20 *
TB6.E2-30.17 *
TB6.E2-30.17 *
TB6.E2-30.18 *
TB6.M(PLUS)-30.24 *
TB6.M(PLUS)-30.24 *
TB6.M(PLUS)-30.25 *
TB6.M(MINUS)-30.29 *
TB6.M(MINUS)-30.30 *
TB6.M(MINUS)-30.32 *
TB6.N(PLUS)-02.28 *
TB6.N(PLUS)-02.28 *

TX1.X1-02.30 *
TX1.X1-02.30 *
TX1.X1-02.30 *
TX1.X2-02.29 *
TX1.X2-02.29 *
TX1.X2-02.29 *

FS RES.1-02.28
FS RES.2-02.22

### #24 RED & WHITE TWISTED

RI.VF(PLUS)-02.09 — W
RI.VF(MINUS)-02.11 — R

RI.V(PLUS)-2PL.07 — W
RI.V(MINUS)-2PL.08 — R

### PLUG WIRING

### #24 WHITE

1PL.01-02.20
1PL.02-04.12
1PL.03-04.15
1PL.04-02.20
1PL.05-04.14
1PL.06-04.15
1PL.09-02.20
1PL.10-04.03
1PL.11-04.16
1PL.12-02.21
1PL.13-04.09
1PL.14-04.16
1PL.15-02.19
1PL.16-05.12
1PL.17-05.15
1PL.18-02.19
1PL.19-05.14
1PL.20-05.15
1PL.21-02.19
1PL.22-05.03
1PL.23-05.16
1PL.26.02.21
1PL.27-05.09
1PL.28-05.16

2PL.01-02.18
2PL.02-06.12
2PL.03-06 15
2PL.04-02.18
2PL.05-06.14
2PL.06-06.15
2PL.09-24.30
2PL.10-06.03
2PL.11-06.16
2PL.12-02.21
2PL.13-06.09
2PL.14-06.16
2PL.15-07.06
2PL.16-02.32
2PL.17-04.23
2PL.18-04.07
2PL.19-13.23
2PL.20-13.15
2PL.21-05.23
2PL.22-05.07
2PL.23-13.20
2PL.24-13.15
2PL.25-06.23
2PL.26-06.07
2PL.27-13.14
2PL.28-13.15

### MOVEABLE JUMPER CONNECTIONS

THE FIRST WIRE CONNECTION NUMBER IS THE PERMANENT ONE. THE SECOND WIRE CONNECTION IS THE MOVEABLE ONE.

#### #24 WHITE

24.28-24.28

THE FIRST WIRE CONNECTION IS THE PERMANENT ONE USING A TERMIPOINT CONNECTION.
THE SECOND WIRE CONNECTION IS THE MOVEABLE ONE USING A SPADE TERMINAL

#### #24 RED

TB3.100-TB3.60
TB3.BFX-TB3.01
TB3.75-TB3.75
TB3.FT-TB3.FT
} WIRES SHOULD BE 3 5 INCHES IN LENGTH

■ REMOVE INDICATED CONNECTIONS FOR G02

FRONT VIEW OF CARD RACK CLOSED POSITION

| | 23 | 24 | 25 | 27 | 30 | ← LOCATION NUMBERS |
|---|---|---|---|---|---|---|
| | 193X188AA G01 (SEE NOTE 5) | 193X188AA 601 | 193X184AA 601 | 193X257AA601 | 193X191AB601 | |
| | | TEST B | MONI TOR | +20VPS | +5VPS | |

| 2 | | 4 | 5 | 6 | 7 | 8 | 10 | 11 | 13 | 15 | ← LOCATION NUMBERS |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 193X190BA602 | | 193X 369AA G01 | 193X 369AA G01 | 193X 369AA G01 | 193X 179AC G01 | 193X 197AA G01 | 193X185AA601 | 193X 195AA G01 | 193X198AA601 | 193X371AA601 | |
| ISOL/PS | | COMM A | COMM B | COMM C | PRO TECT ION | MODU LA TOR | REV LOGIC | DIV IDER | SYN CHRONIZER II | REGULATOR | |

NOTES

1. LOCATION NUMBERS AT THE TOP OF EACH CARD INDICATE THE SPACE NUMBER THAT THE CARD'S RECEPTACLE OCCUPIES. THE CARD OCCUPIES THAT SAME SPACE AND AN INTEGRAL NUMBER OF SPACES TO THE LEFT. THE SPACE UNIT REQUIREMENT FOR A CARD CAN BE DETERMINED BY SUBTRACTING THE SPACE NUMBER OF THE CARD TO THE LEFT.

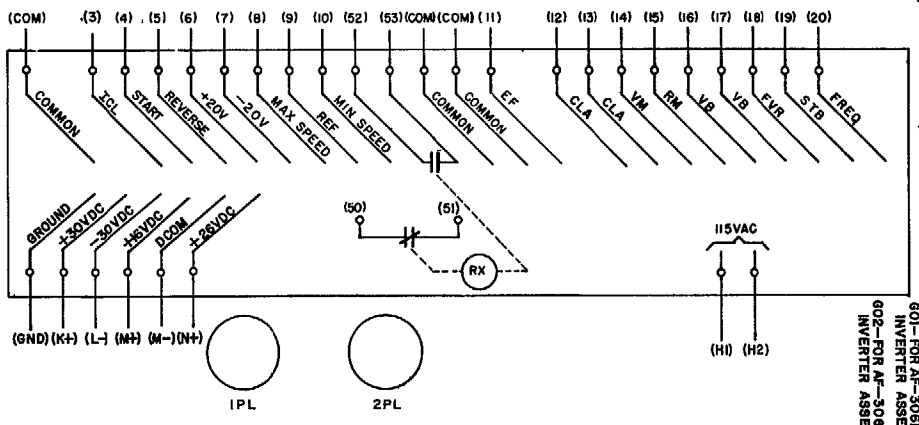
2. THE WIRE TABLE LISTS ALL WIRE CONNECTIONS IN THE DRIVER. A CONNECTION IS INDICATED BY A CARD LOCATION NUMBER, AND THE RECEPTACLE PIN NUMBER SEPARATED BY A PERIOD (E.G. 02.30 INDICATES LOCATION 2, PIN 30). OTHER CONNECTIONS SUCH AS TO A TERMINAL BOARD ARE INDICATED BY THE DEVICE NOMENCLATURE AND A POINT NUMBER (E.G. TB.01). A WIRE BETWEEN TWO POINTS IS INDICATED BY A DASH BETWEEN THE TWO CONNECTION POINT NOMENCLATURES (E.G. TB.21 - 23.06). A CONNECTION MADE BY BUS BAR AS SHOWN IN SKETCH "B" IS INDICATED IN THE TABLE AT THE BEGINNING OF THE WIRE TABLE.

3. THE RECEPTACLE IS NUMBERED CORRESPONDING TO ITS INDIVIDUAL PINS. CARDS HAVE TAB NUMBERS WHICH CORRESPOND TO PIN NUMBERS ON THE RECEPTACLE. FOR A CARD WITH TABS ON BOTH SIDES THE CORRESPONDING RECEPTACLE HAS 64 INDIVIDUAL PINS. A CARD WITH TABS ON ONE SIDE HAS A CORRESPONDING RECEPTACLE WITH 32 PINS.

4. ON PRINTED CIRCUIT CARDS USED IN THIS REGULATOR THE LETTERS "AA" AFTER THE BASIC CARD CATALOG NUMBER INDICATES ORIGINAL DESIGN. SUBSEQUENT DESIGNS WITH THE SAME BASIC NUMBERS AND GROUP NUMBER WITH THE SECOND LETTER CHANGED, SUCH AS AB, AC, AD, ETC., ARE DIRECTLY INTERCHANGEABLE AND MAY BE SUPPLIED IN PLACE OF THE "AA" CARD. AN EARLIER DESIGN CANNOT REPLACE A LATER DESIGN, E.G. AN AA CANNOT REPLACE AN AB.

5. LOCATION NUMBER 24 HAS NO PERMANENT CARD, IT IS A TEST RECEPTACLE AND MAY BE USED WITH THE TEST CARD IN LOCATION NUMBER 23.

DRIVER SUBASSEMBLY EXTERNAL CONNECTIONS




G01- FOR AF-306I INVERTER ASSEMBLIES
G02—FOR AF-3060 INVERTER ASSEMBLIES

33IX398BA601,G02

FIRST MADE FOR STANDARD LINE

TITLE CONNECTION DIAGRAM
AF-3062 DRIVER

GENERAL ELECTRIC

36C764839BA
CONT ON SHEET 3

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES: 718A505
SURFACES: √
FRACTIONS
TOLERANCES ON MACHINED DIMENSIONS: DECIMALS ± , ANGLES ±

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

| REVISIONS | | | |
|---|---|---|---|
| REV NO | DESCRIPTION | APPROVALS | DATE |
| 0 | PRINTED FOR FIRST ISSUE | [illegible] | JUNE 24, 1976 |

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SPEED VARIATOR
ERIE, PA
DIV OR LOCATION
CONT ON SHEET 3

36C764839BA
SH NO 2

PRINTS TO

| | | | | | | |
|---|---|---|---|---|---|---|
| 5T5(5D) | 50C(BK) | 50C(BK) | 5L(BW)15 | 50(BK) | AW (BW) | 5J(CD) 51(I) |
| | | | | | | 98(13)23 |

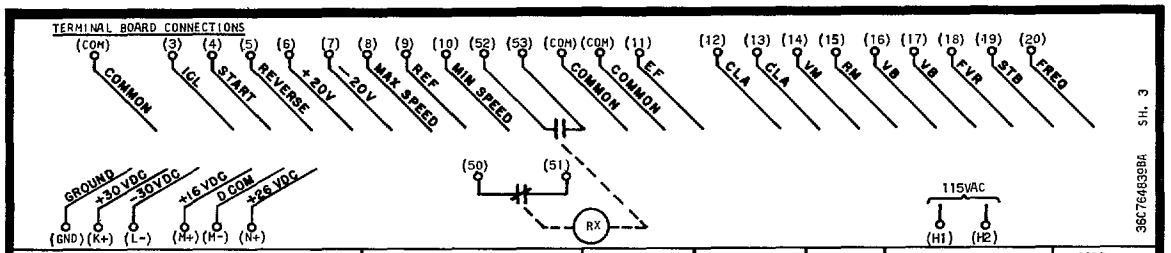
GENERAL ELECTRIC | 36C764839BA | CONT ON SHEET FL | SH NO 3

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE LABEL ARTWORK & NOTES
AF 3062 DRIVER

FIRST MADE FOR STANDARD LINE

331X358BAG01, G02




## TEST CARD

THE TEST CARD MAY BE PLACED IN EITHER RECEPTACLE A OR B A MOVABLE JUMPER CONNECTED TO PIN 28 OF THE TEST RECEPTACLES MAY BE USED TO MONITOR ANY POINT IN THE BACK PLANE THROUGH THE "SEL" TEST CONNECTOR ON THE TEST CARDS

### TEST POINTS

| | A | B | |
|---|---|---|---|
| 1 | REF | CFSP | A |
| 2 | MV | IFSP | A |
| 3 | 1FP | CFSN | A |
| 4 | FXL | IFSN | A |
| 5 | EF | CFSP | B |
| 6 | QFP/4 | IFSP | B |
| 7 | 1FFY | CFSN | B |
| 8 | 1FFZ | IFSN | B |
| 9 | OMPA | CFSP | C |
| 10 | OMPB | IFSP | C |
| 11 | OMPC | CFSN | C |
| 12 | TW | IFSN | C |
| 13 | OFT | CL | |
| 14 | OSTART | CSA | |
| 15 | OMV | CSB | |
| 16 | ORUN | CSC | |
| 17 | 1FFXP | CLA | |
| 18 | OSI | VFB | |
| 19 | OREV | FV | |

## MONITOR CARD

| | |
|---|---|
| +20V | DRIVER TRIP (IOC WILL ALSO LIGHT) |
| DFP | INSTANTANEOUS OVER-CURRENT TRIP |
| SEL | CURRENT LIMIT ON |
| −5V | FREQUENCY SIGNAL |
| +5V | FILTER F.B |
| −20V | CCA |
| | CCB |
| | CCC |
| | 1FFA |
| | COMMON |
| READ | |
| COM | |

19

## REVERSE LOGIC

P475 SLIP COMP
P476 I.R COMP
P477 TIME UP
P478 TIME DOWN

## REGULATOR

P265 VOLTAGE BOOST
P267 V/HZ
P266 CURR LIMIT
P268 MIN FREQ

| SPACE NUMBER | NAME | CARD NUMBER |
|---|---|---|
| 2 | ISOLATOR/PS | 193X190BAG02 |
| 4 | COMMUTATION "A" | 193X369ABG01 |
| 5 | COMMUTATION "B" | 193X369ABG01 |
| 6 | COMMUTATION "C" | 193X369ABG01 |
| 7 | DRIVE PROTECTION | 193X179ACG01 |
| 8 | MODULATOR | 193X197AAG01 |
| 10 | REVERSE LOGIC | 193X185AAG01 |
| 11 | DIVIDER | 193X195ABG01 |
| 13 | SYNCHRONIZER II | 193X198AAG01 |
| 15 | REGULATOR | 193X371ABG01 |
| 23 | TEST A | 193X188AAG01 |
| 24 | TEST B | |
| 25 | MONITOR | 193X184AAG01 |
| 27 | +20V PS | 193X257AAG01 |
| 30 | +5V PS | 193X191ABG01 |

## BASE FREQUENCY SELECTION

BASE FREQUENCY IS SELECTED BY TWO JUMPERS ON TB3 AS FOLLOWS

## POWER SUPPLY CARDS

BEFORE REPLACING FUSES(S) ON POWER SUPPLY CARDS, CHECK FOR POSSIBLE SHORT(S) ACROSS THE OUTPUT(S). REPLACE WITH THE SAME TYPE AND RATING (SPARE FUSE MOUNTED ON THE CARD)

SPACE FOR CAUTION LABEL 104X905BA148

## FREQUENCY TRIP SELECTION

TRIP FREQUENCY IS SELECTED BY A JUMPER ON TB3 AS FOLLOWS

| TB3 (7S) JUMPER | 75 | 110 | 150 | 200 | 250 |
|---|---|---|---|---|---|
| FREQUENCY TRIP (HZ) | 75 | 110 | 150 | 200 | 250 |
| TB3(FT) JUMPER TO (X4) | 300 | 440 | 600 | 800 | 1000 |

SPACE FOR WARNING LABEL NP104X905BA068

**601, 602 DRIVER**

| MULTIPLIER: TB3 JUMPER / BFX (100) | HERTZ 40 | 50 | 60 | 70 | 80 | 100 |
|---|---|---|---|---|---|---|
| (1) | 20 | 25 | 30 | 35 | 40 | 50 |
| (2) | 40 | 50 | 60 | 70 | 80 | 100 |
| (4) | 80 | 100 | 120 | 140 | 160 | |

SPACE FOR CAUTION LABEL NP 104X905BA067

### DRIVER DOOR LABEL

1. SIZE AS INDICATED.
2. PRINT ON FASSON WHITE CONVENTION BADGE (LAPEL) STOCK NO 65P.
3. CUT TO SIZE.

11"

13"

DRAWING NO | SH NO | CONT ON SHEET | REV NO Ø 1

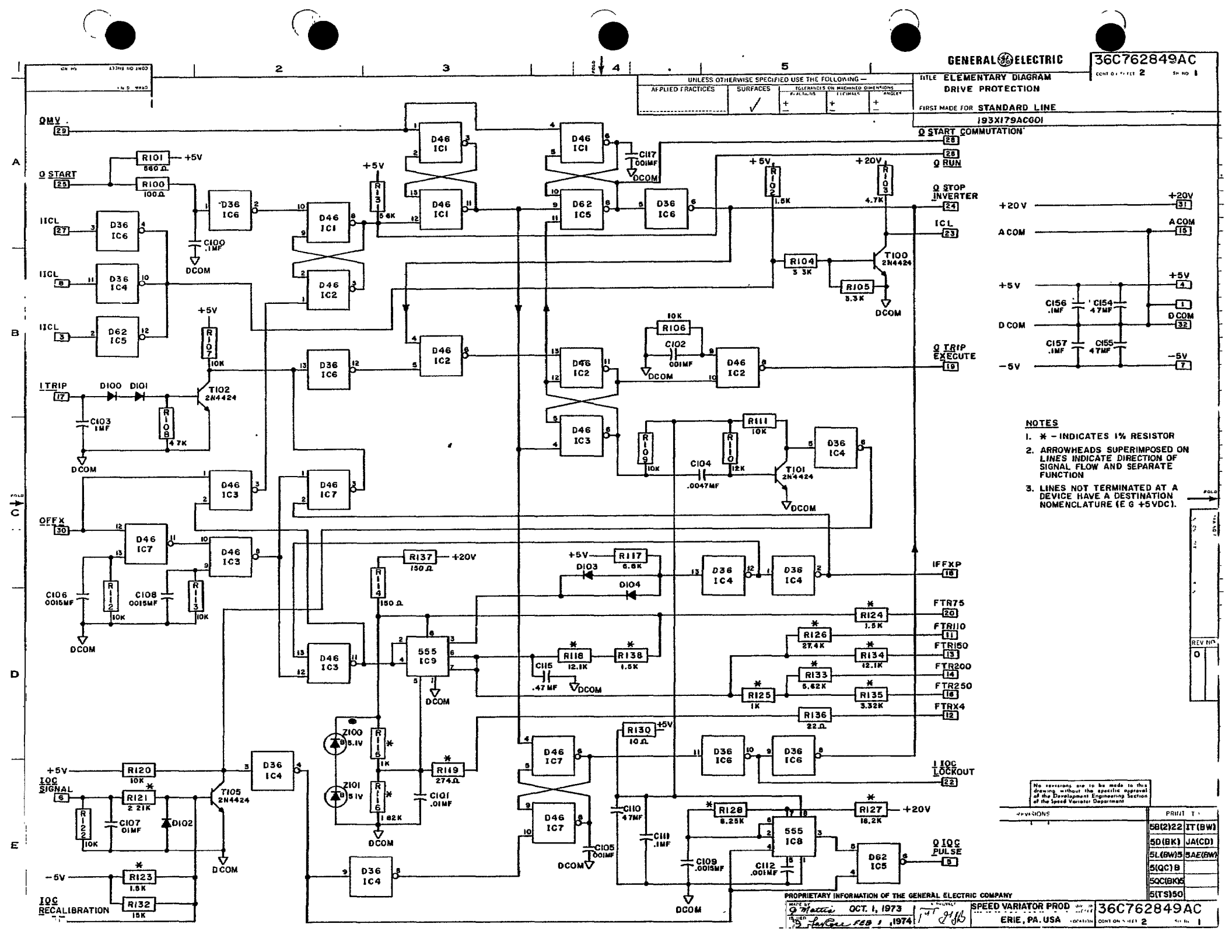
No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

REVISIONS
1 M D Pogle SEPT 2 1976 RV67A598

PRINTS TO

| | |
|---|---|
| 5(S) | JA(CO) |
| 5B(3)23 | AW(BW) |
| 5D(BK) | 5L(BWS) |
| 5QC(B) | |
| 5QC(BK)S | |
| 5(TS)50 | |

MADE BY M D Pogle JUNE 24, 1976
TRACED M D Pogle AUG 12, 1976
APPROVALS CEK
SPEED VARIATOR DIV OR DEPT ERIE, PA. LOCATION
36C764839BA CONT ON SHEET FL SH NO 3


GENERAL ELECTRIC
36C762849AC
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
TITLE ELEMENTARY DIAGRAM
DRIVE PROTECTION
FIRST MADE FOR STANDARD LINE
193X179ACG01
Q START COMMUTATION
Q RUN
Q STOP INVERTER
ICL
Q TRIP EXECUTE
IFFXP
FTR75
FTR110
FTR150
FTR200
FTR250
FTRX4
I IOC LOCKOUT
Q IOC PULSE
+20V
ACOM
+5V
DCOM
-5V
NOTES
1. * - INDICATES 1% RESISTOR
2. ARROWHEADS SUPERIMPOSED ON LINES INDICATE DIRECTION OF SIGNAL FLOW AND SEPARATE FUNCTION
3. LINES NOT TERMINATED AT A DEVICE HAVE A DESTINATION NOMENCLATURE (E G +5VDC).
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY
SPEED VARIATOR PROD
ERIE, PA. USA
36C762849AC

GENERAL ELECTRIC

36C762849AG

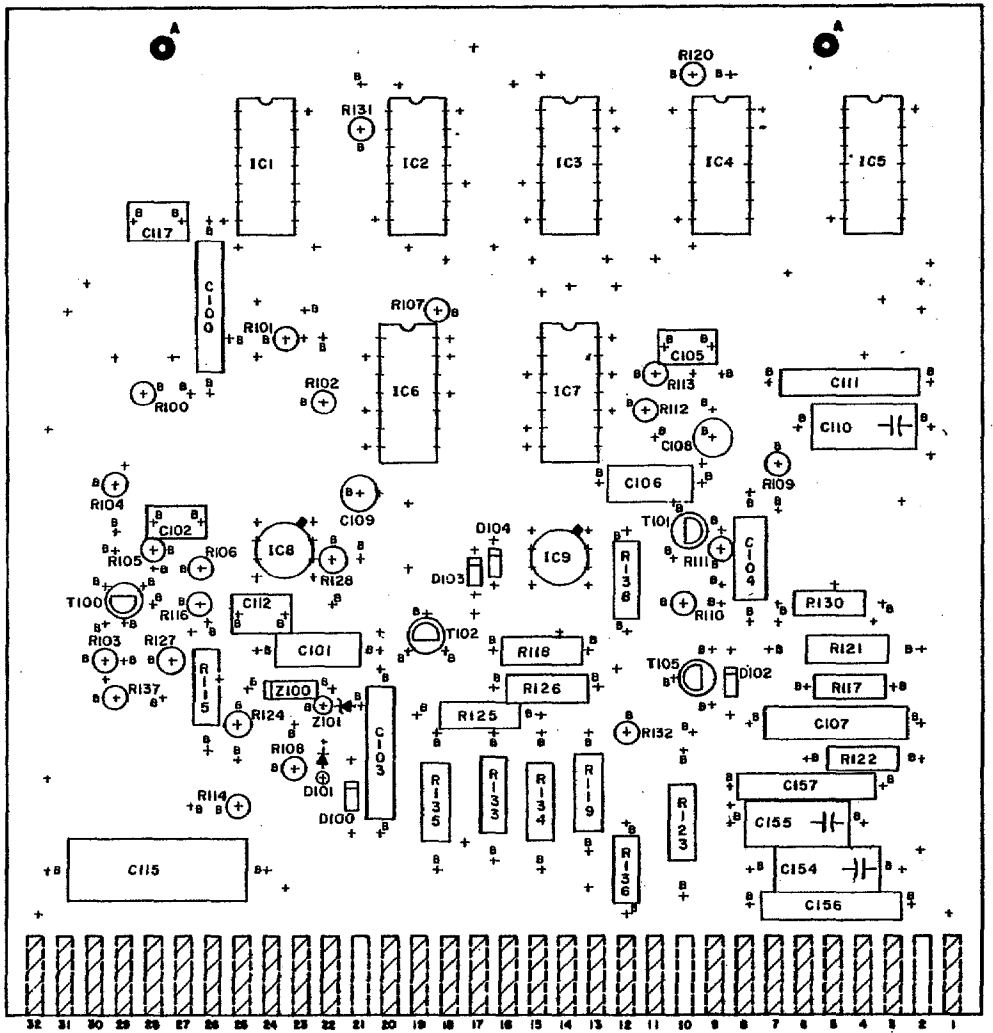
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:
APPLIED PRACTICES: 710A816 | SURFACES: ✓ | TOLERANCES ON INCLINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE PRINTED CIRCUIT DIAGRAM
DRIVE PROTECTION
FIRST MADE FOR STANDARD LINE
193X179ACG01

## HOLE TABULATION

ALL HOLES .032 DIA. EXCEPT THE HOLES TABULATED BELOW

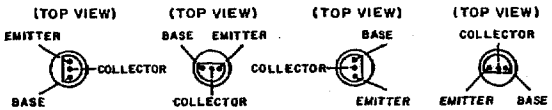
| LOC. | DIA. | QUAN. |
|---|---|---|
| A | .157 | 2 |
| B | .040 | 130 |

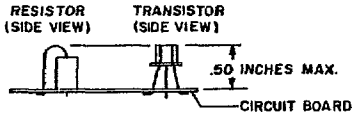


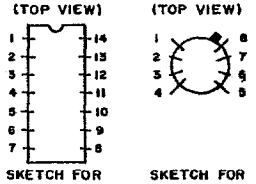

### NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $8.500^{+.000}_{-.018}$ X $6.130^{+.002}_{-.008}$
4. ALL TRANSISTORS AND RESISTORS SHALL BE MOUNTED TO A .50 INCH MAXIMUM ABOVE THE CARD SURFACE AS SHOWN BELOW.

### TRANSISTOR LEAD SKETCHES




| | | | | |
|---|---|---|---|---|
| 2X SCALE | 36A352983CA005 REAR ETCHED BOARD DWG. | 36A352982CA005 FRONT ETCHED BOARD DWG. | 1 FRONT SPACE | 0 BACK UNIT REQUIREMENT |

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

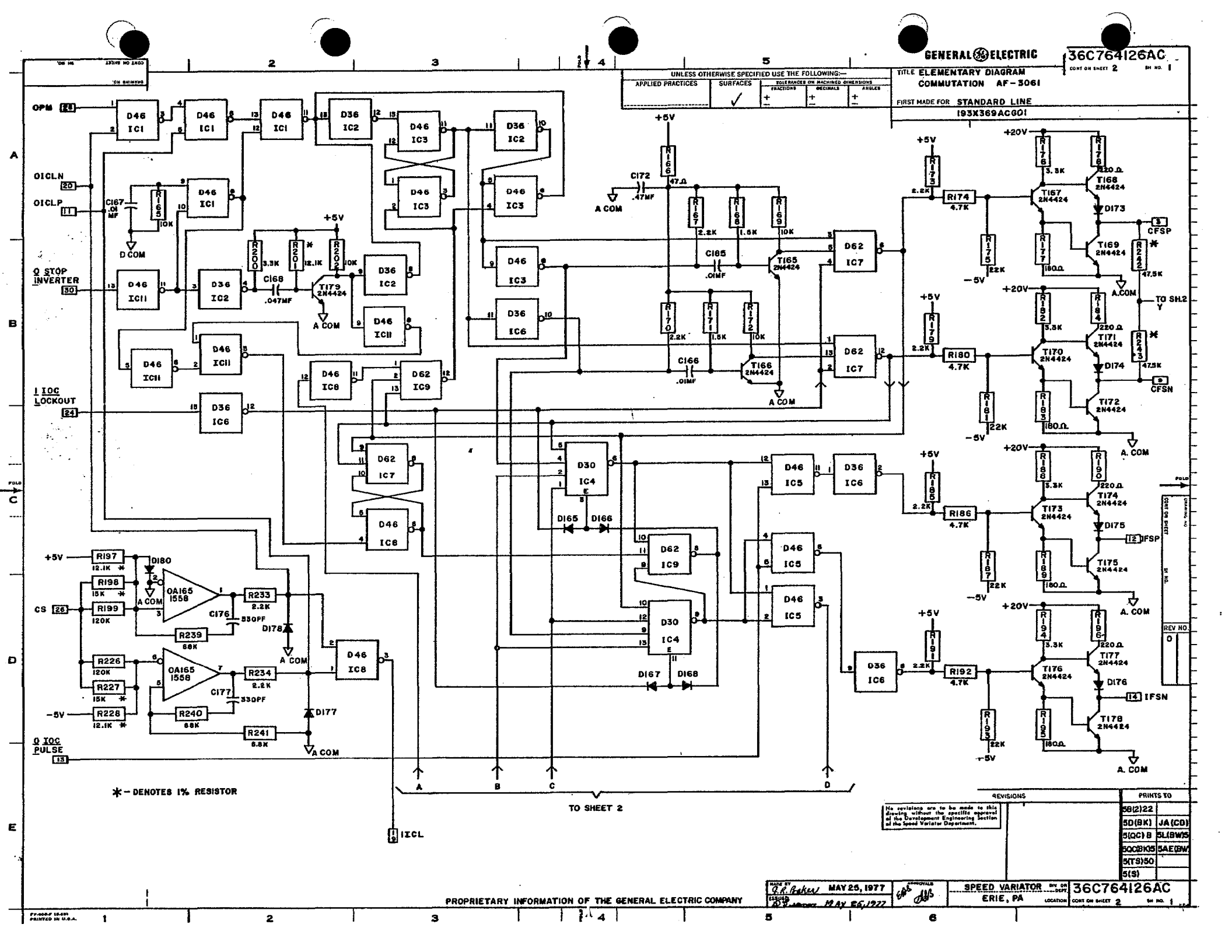
PRINTS TO

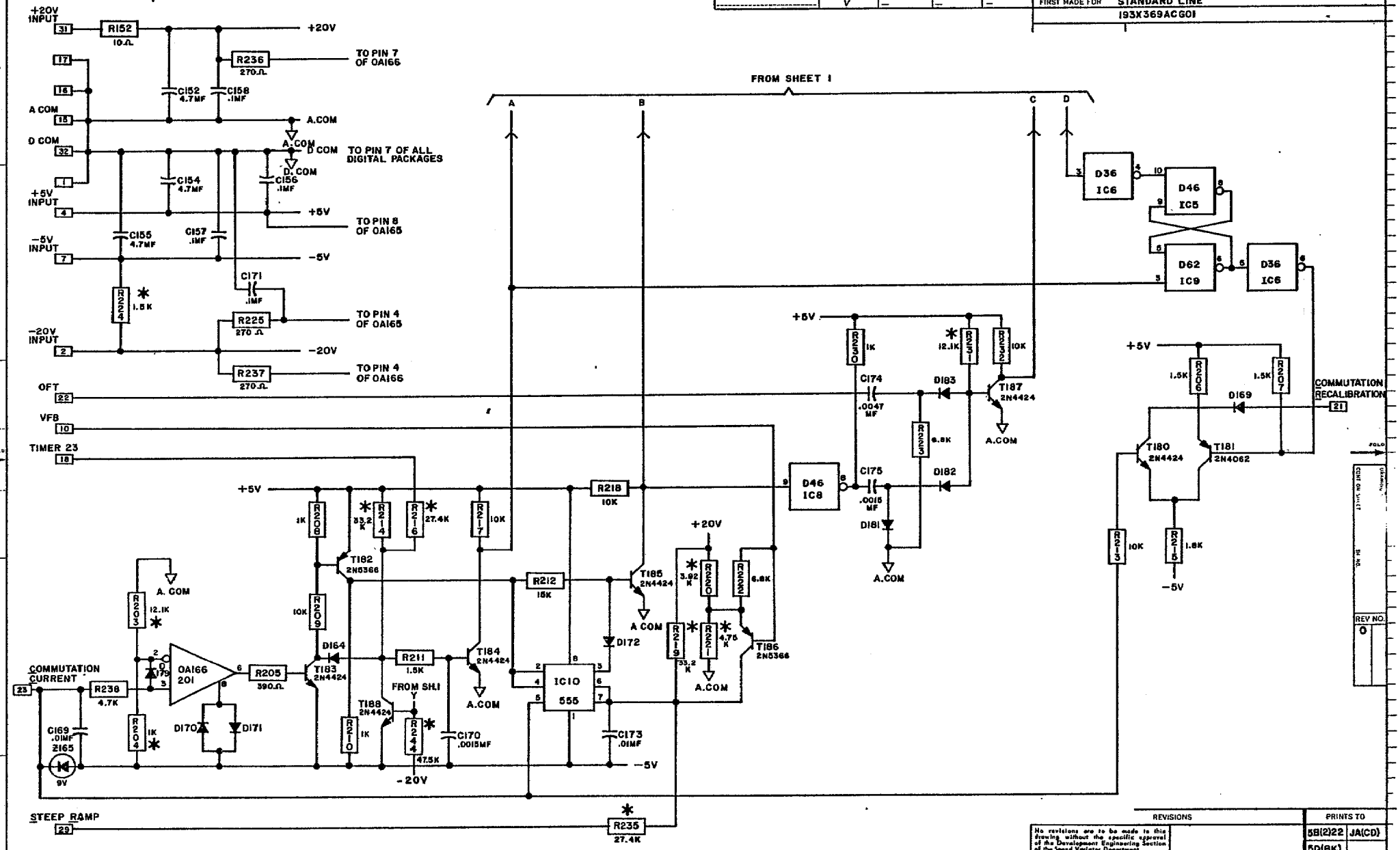
| | |
|---|---|
| 5QC(B) | 5 (S) |
| 5B(2)22 | 5D(BK) |
| 5L(BW)5 | 5B (S) |
| 5QC(BK)5 | 5AE(BW) |
| 5(TS)50 | JA(CD) |
| 1T(BW) | |

REV NO. G

| | | | | |
|---|---|---|---|---|
| DRAWN BY G. Martin | OCT. 3, 1973 | APPROVALS | SPEED VARIATOR PROD DEPT. ERIE, PA. USA LOCATION | 36C762849AC CONT ON SHEET FL SH NO 2 |

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY


GENERAL ELECTRIC
36C764126AC
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES
SURFACES
TOLERANCES ON MACHINED DIMENSIONS
FRACTIONS
DECIMALS
ANGLES
TITLE ELEMENTARY DIAGRAM
COMMUTATION AF-3061
FIRST MADE FOR STANDARD LINE
193X369ACG01
OPM 28
OICLN 20
OICLP 11
Q STOP INVERTER 30
I IOC LOCKOUT 24
CS 26
Q IOC PULSE 13
IICL 9
D COM
A COM
-5V
+5V
+20V
TO SH.2
CFSP
CFSN
IFSP
IFSN
A. COM
TO SHEET 2
* — DENOTES 1% RESISTOR
REVISIONS
No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.
PRINTS TO
58(2)22
5D(BK) JA(CD)
5(QC) B 5L(BW)S
5QCBK)S 5AE(BW
5(TS)50
5(S)
MADE BY R.E. Baker MAY 25, 1977
ISSUED MAY 26, 1977
SPEED VARIATOR ERIE, PA
36C764126AC
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY




GENERAL ELECTRIC

36C764126AC CONT ON SHEET 3 SH NO. 2

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

| APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ± |
|---|---|---|---|---|
| | ✓ | | | |

TITLE ELEMENTARY DIAGRAMS
COMMUTATION AF-3061

FIRST MADE FOR STANDARD LINE
193X369ACG01

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

REV NO. 0

| PRINTS TO | |
|---|---|
| 5B(2)22 | JA(CD) |
| 5D(8K) | |
| 5(QC)8 | 5L(BW)5 |
| 5QC(8K)5 | 5AE(BW) |
| 5(TS)50 | |
| 5(S) | |

MADE BY A.R. Baker MAY 25, 1977
ENGINEERING APPROVALS
speed variator DIV. OR DEPT.
ERIE, PA. LOCATION
36C764126AC CONT ON SHEET 3 SH NO. 2

PREPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

PP-808-P 14-72 PRINTED IN U.S.A.

GENERAL ELECTRIC

36C764126AC
CONT ON SHEET FL. SH NO 3

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING -

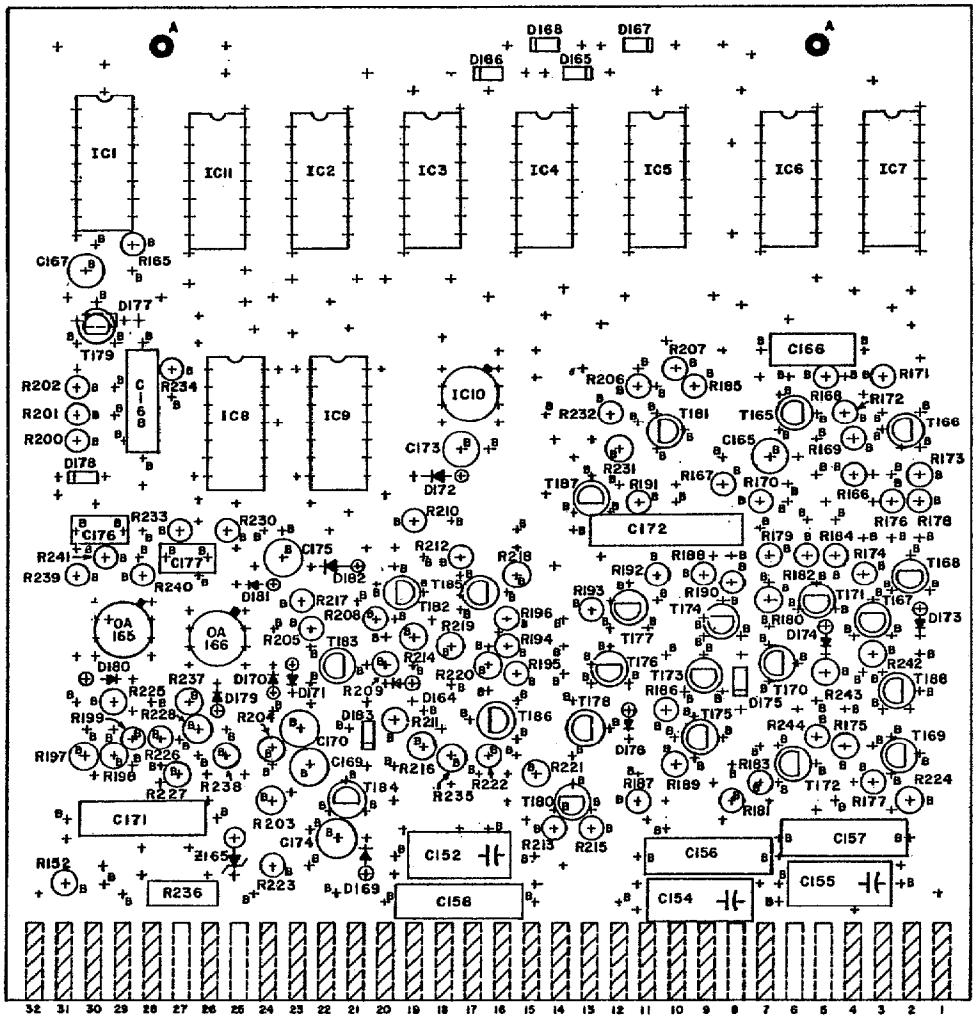
| APPLIED PRACTICES | SURFACES | TOLERANCES OR MACHINED DIMENSIONS FRACTIONS | DECIMALS | ANGLES |
|---|---|---|---|---|
| 710A515 | ✓ | ± | ± | ± |

TITLE PRINTED CIRCUIT DIAGRAM
COMMUTATION AF-3061
FIRST MADE FOR STANDARD LINE
193X369ACG01

## HOLE TABULATION

ALL HOLES .032 DIA. EXCEPT THE HOLES TABULATED BELOW

| LOC. | DIA. | QUAN. |
|---|---|---|
| A | .157 | 2 |
| B | .040 | 269 |

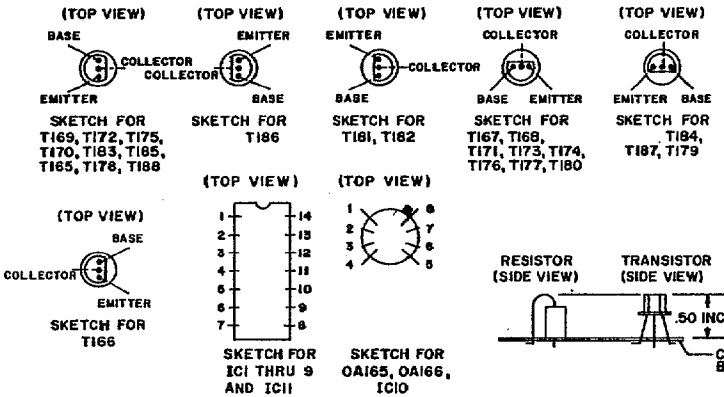


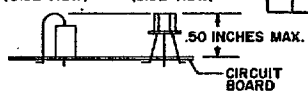

SEE NOTE 1 & 2

## NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$

## TRANSISTOR LEAD SKETCHES




| 2 X SCALE | 36A353788AB005 REAR ETCHED BOARD DWG. | 36A353789AB005 FRONT ETCHED BOARD DWG. | 1 FRONT SPACE UNIT REQUIREMENT | 0 BACK |
|---|---|---|---|---|

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

REV NO. 0

PRINTS TO

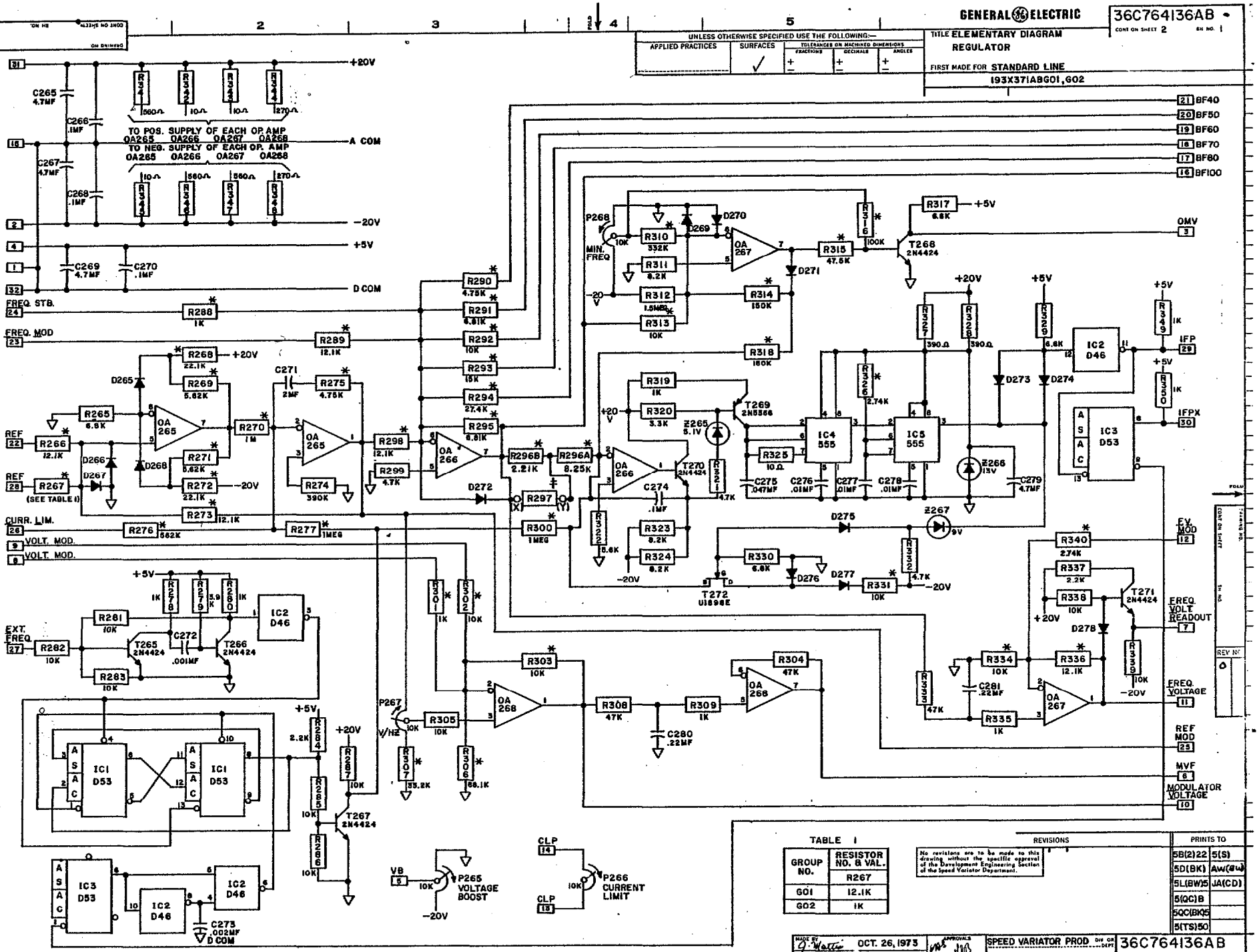
| | |
|---|---|
| 5B(2)22 | |
| 5D(BK) | JA(CD) |
| 5(QC)B | 5L(BW)5 |
| 500(BK)5 | 5AE(BW) |
| 5(TS)50 | |
| 5(S) | |

MADE BY J.R. Baker MAY 25, 1977
ISSUED J.R. Baker MAY 25, 1977
APPROVALS

SPEED VARIATOR PROD SYS OR DEPT ERIE, PA. LOCATION

36C764126AC
CONT ON SHEET FL. SH NO 3

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY


GENERAL ELECTRIC
36C764136AB
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES
SURFACES
TOLERANCES ON MACHINED DIMENSIONS
FRACTIONS
DECIMALS
ANGLES
TITLE ELEMENTARY DIAGRAM
REGULATOR
FIRST MADE FOR STANDARD LINE
193X371A8G01,G02
+20V
A COM
-20V
+5V
D COM
FREQ. STB.
FREQ. MOD
REF
CURR. LIM.
VOLT. MOD.
EXT. FREQ.
TO POS. SUPPLY OF EACH OP AMP
0A265 0A266 0A267 0A268
TO NEG. SUPPLY OF EACH OP. AMP
0A265 0A266 0A267 0A268
BF40
BF50
BF60
BF70
BF80
BF100
OMV
IFP
IFPX
F.V. MOD
FREQ. VOLT READOUT
FREQ. VOLTAGE
REF MOD
MVF
MODULATOR VOLTAGE
P265 VOLTAGE BOOST
P266 CURRENT LIMIT
TABLE I
GROUP NO.
RESISTOR NO. & VAL.
R267
G01
12.1K
G02
1K
REVISIONS
No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.
PRINTS TO
5B(2)22 5(S)
5D(BK) AW(BW)
5L(BW)5 JA(CD)
5(QC)B
5QC(BK)5
5(TS)50
MADE BY O. Watts OCT. 26, 1973
ISSUED Carl Marsh Dec. 3, 1975
APPROVALS
SPEED VARIATOR PROD. DIV. OR DEPT.
ERIE, PA. USA LOCATION
36C764136AB
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

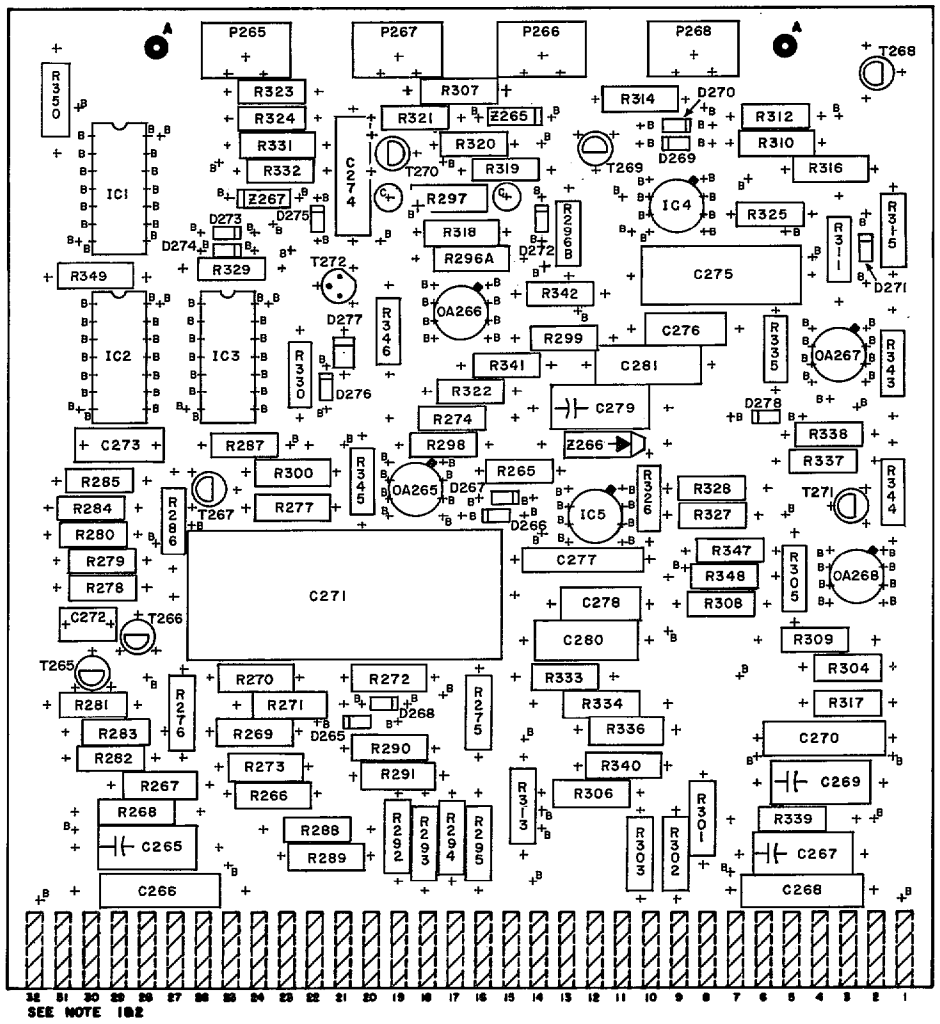
GENERAL ELECTRIC

36C764136AB
CONT ON SHEET FL. SH NO 2

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING -

| APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS FRACTIONS | DECIMALS | ANGLES |
|---|---|---|---|---|
| 710A815 | ✓ | ± | ± | ± |

TITLE PRINTED CIRCUIT DIAGRAM
REGULATOR CARD
FIRST MADE FOR STANDARD LINE AF-3061
193X371ABG01,G02




**HOLE TABULATION**

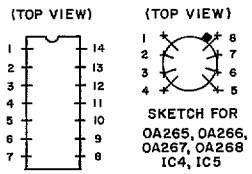
ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

| LOC. | DIA. | QUAN. |
|---|---|---|
| A | .157 | 2 |
| B | .032 | 168 |
| C | .081 | 2 |

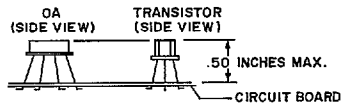
NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$
4. ALL TRANSISTORS AND OP AMPS SHALL BE MOUNTED TO A .50 INCH MAXIMUM ABOVE THE CARD SURFACE AS SHOWN BELOW.
5. FOR-G02 ONLY
   STAMP ADHESIVE LABEL-G02. ADD LABEL AS SUFFIX TO 193X371AA IN PLACE OF G01 STAMPED ON PRINTED CIRCUIT BOARD.

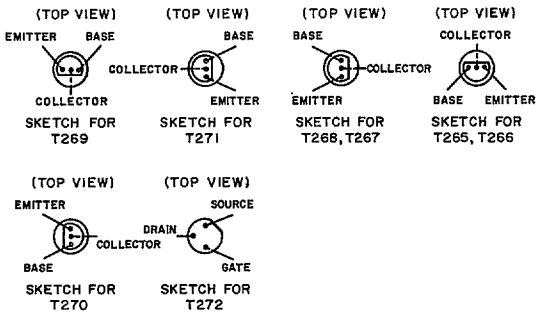
IC & OA LEAD SKETCHES




COMPONENT HEIGHT SKETCH




TRANSISTOR LEAD SKETCHES




| 2X | 36A353697AC005 | 36A353698AC005 | I | O |
|---|---|---|---|---|
| SCALE | REAR ETCHED BOARD DWG. | FRONT ETCHED BOARD DWG. | FRONT SPACE | BACK UNIT REQUIREMENT |

REVISIONS

| REV NO. | | | |
|---|---|---|---|
| 1 | M. Cotty Nov. 30, 1978 | PER SV-883#37 | |

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

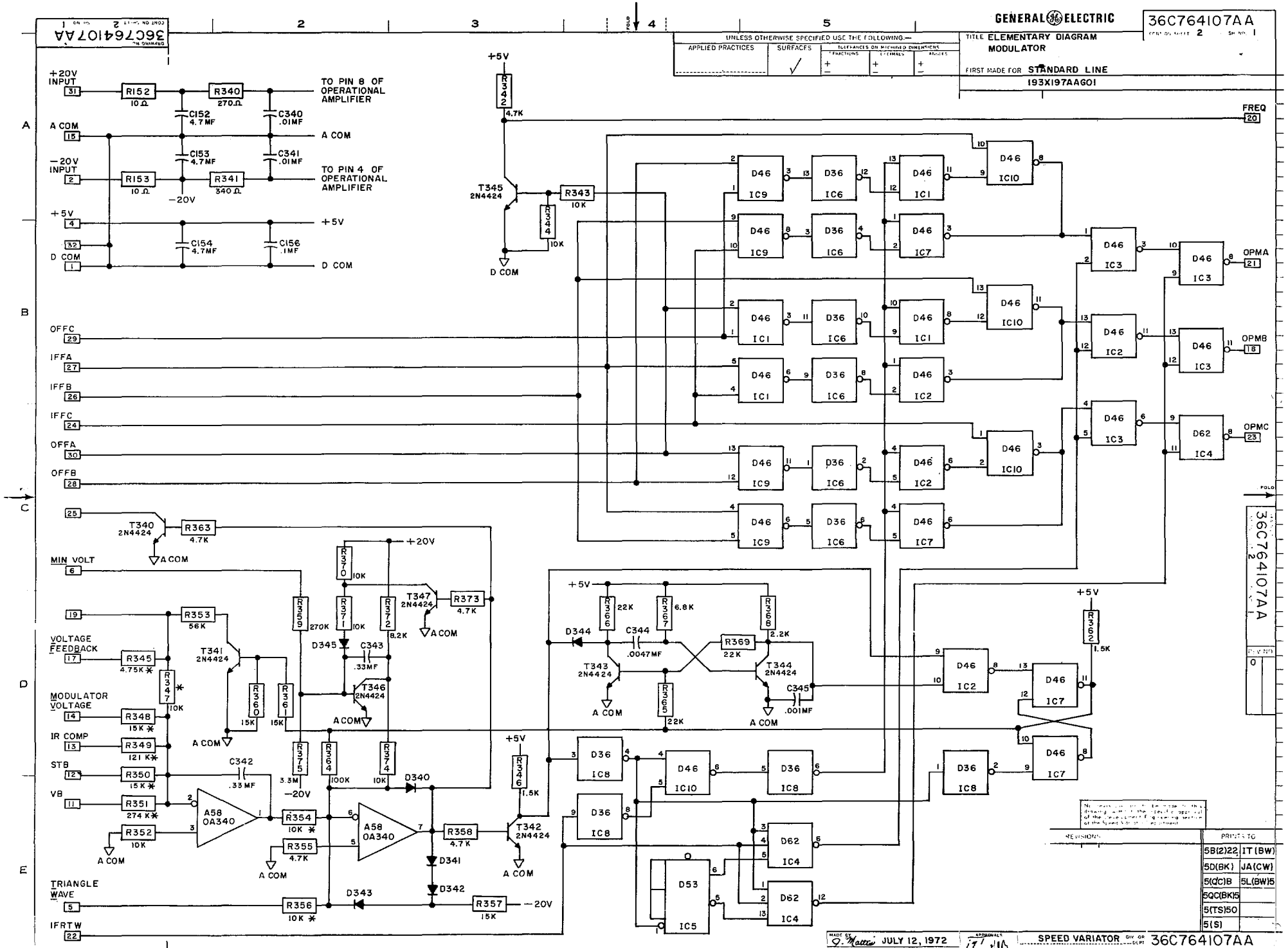
PRINTS TO

| | |
|---|---|
| 5B(8)M | 5J(BW) |
| 5R(2)BW | TS(50) |
| 5L(2)BW | |
| AW(BW) | |
| 5D(CD) | |
| JA(CD) | |

DRAWN BY G. Mattia NOV. 21, 1973
ISSUED M. Cotty Nov. 30, 1978
APPROVALS ERS JJB
SPEED VARIATOR PROD. DIV. OR DEPT. ERIE, PA. USA LOCATION

36C764136AB
CONT ON SHEET FL. SH NO 2

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY


GENERAL ELECTRIC
36C764107AA
TITLE ELEMENTARY DIAGRAM
MODULATOR
FIRST MADE FOR STANDARD LINE
193X197AAG01
+20V INPUT
-20V INPUT
+5V
D COM
A COM
TO PIN 8 OF OPERATIONAL AMPLIFIER
TO PIN 4 OF OPERATIONAL AMPLIFIER
R152 10Ω
R153 10Ω
R340 270Ω
R341 340Ω
C152 4.7MF
C153 4.7MF
C154 4.7MF
C156 .1MF
C340 .01MF
C341 .01MF
FREQ
OPMA
OPMB
OPMC
OFFC
IFFA
IFFB
IFFC
OFFA
OFFB
MIN VOLT
VOLTAGE FEEDBACK
MODULATOR VOLTAGE
IR COMP
STB
VB
TRIANGLE WAVE
IFRTW
T340 2N4424
T341 2N4424
T342 2N4424
T343 2N4424
T344 2N4424
T345 2N4424
T346 2N4424
T347 2N4424
R343 10K
R344 10K
R345 4.75K*
R348 15K*
R349 121K*
R350 15K*
R351 274K*
R352 10K
R354 10K*
R355 4.7K
R356 10K*
R357 15K
R358 4.7K
R359 270K
R360 15K
R361 15K
R363 4.7K
R364 100K
R365 22K
R366 22K
R367 6.8K
R368 2.2K
R369 22K
R370 10K
R371 10K
R372 8.2K
R373 4.7K
R374 10K
R375 3.3M
R376 1.5K
R362 1.5K
C342 .33MF
C343 .33MF
C344 .0047MF
C345 .001MF
D340
D341
D342
D343
D344
A58 0A340
D53 IC5
D36 IC8
D46 IC9
D46 IC1
D46 IC6
D46 IC10
D46 IC2
D46 IC7
D46 IC3
D62 IC4
SPEED VARIATOR
36C764107AA
JULY 12, 1972

GENERAL ELECTRIC

36C764107AA
CONT ON SHEET FL SH NO 2

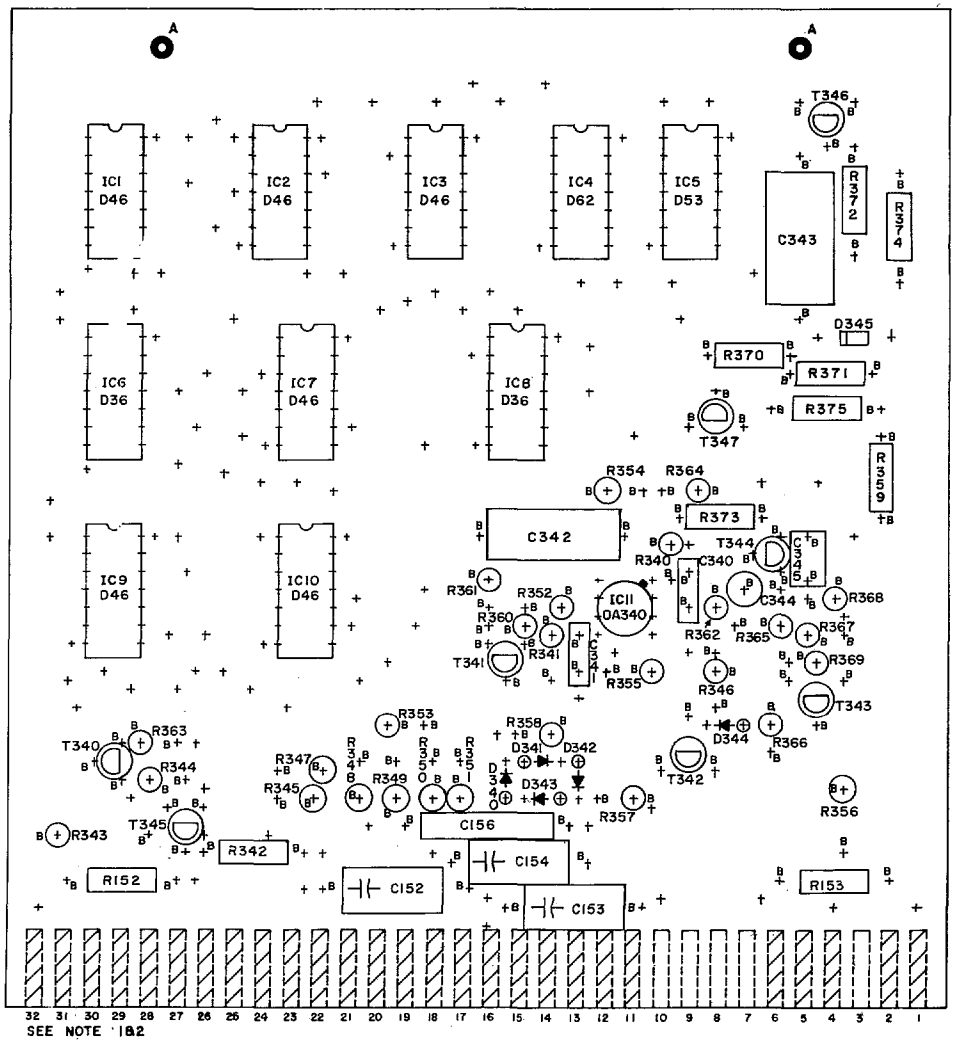
| UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING | | | | |
|---|---|---|---|---|
| APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS | | |
| | | FRACTIONS | DECIMALS | ANGLES |
| 718A515 | ✓ | ± | ± | ± |

TITLE PRINTED CIRCUIT DIAGRAM
MODULATOR
FIRST MADE FOR STANDARD LINE
193X197AAG01

## HOLE TABULATION

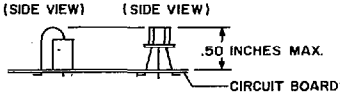
ALL HOLES .032 DIA. EXCEPT THE HOLES TABULATED BELOW

| LOC. | DIA. | QUAN. |
|---|---|---|
| A | .157 | 2 |
| B | .040 | 120 |

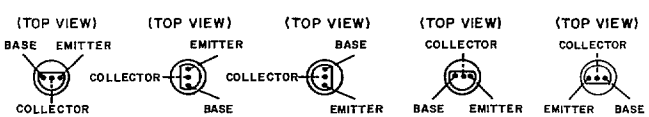



## NOTES

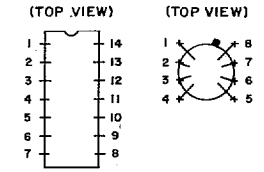
1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$
4. ALL TRANSISTORS AND RESISTORS SHALL BE MOUNTED TO A .50 INCH MAXIMUM ABOVE THE CARD SURFACE AS SHOWN BELOW.




## TRANSISTOR LEAD SKETCHES




## IC LEAD SKETCHES




| | | | | |
|---|---|---|---|---|
| 2X | 36A352956AA005 | 36A352955AA005 | 1 | — |
| SCALE | REAR ETCHED BOARD DWG. | FRONT ETCHED BOARD DWG. | FRONT | BACK |
| | | | SPACE UNIT REQUIREMENT | |

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Varietor Department.

REVISIONS

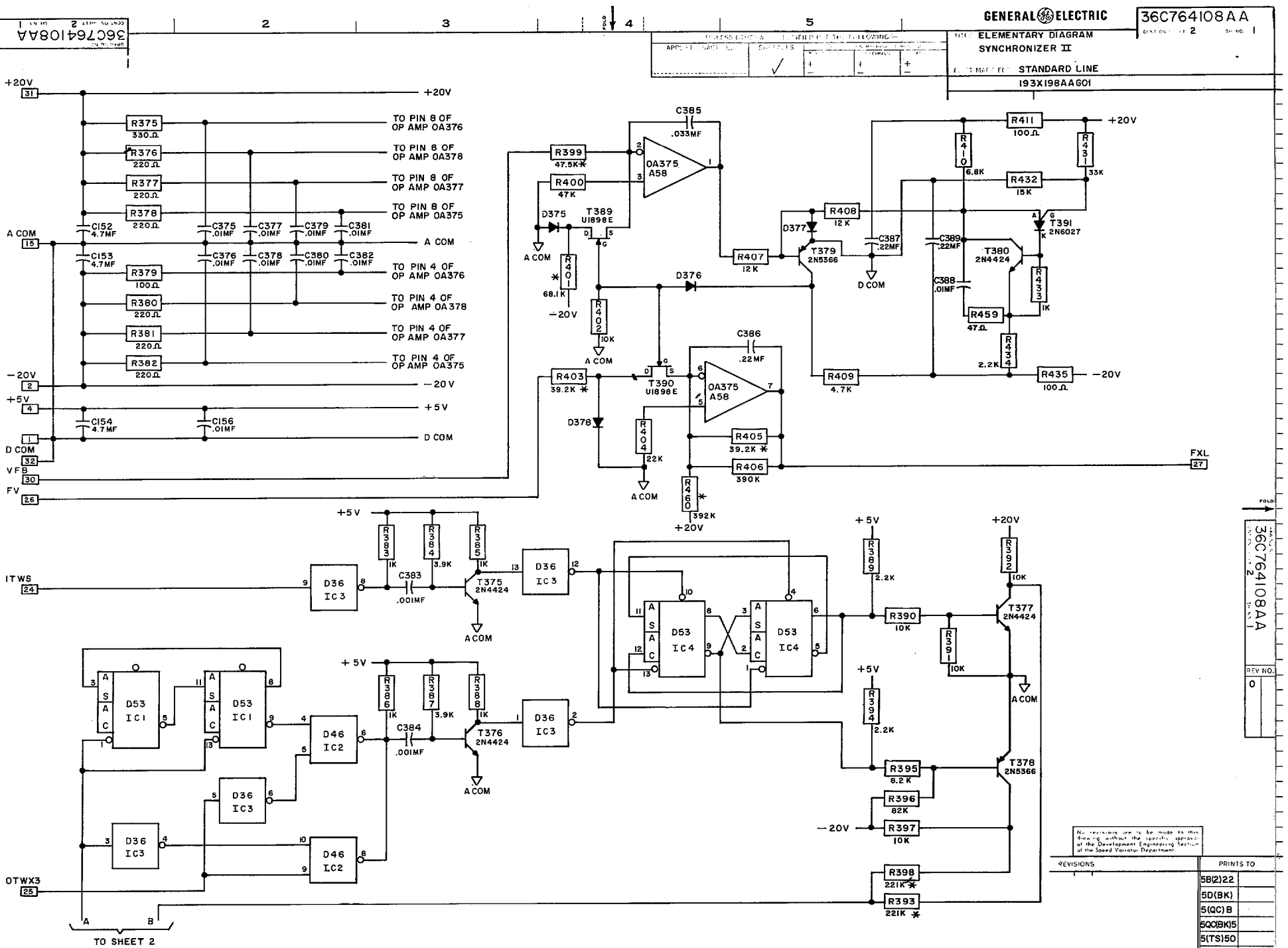
| PRINTS TO | |
|---|---|
| 5B(2)22 | 1T(BW) |
| 5D(BK) | JA(CD) |
| 5(QC)B | 5L(BW)5 |
| 500(BK)5 | |
| 5(TS)50 | |
| 5(S) | |

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

MADE BY P. Matta AUG. 25, 1972
APPROVED
SPEED VARIATOR DIV. OR DEPT.
ERIE, PA. LOCATION
36C764107AA
CONT ON SHEET FL SH NO 2

36C764107AA
CONT ON SHEET FL SH NO 2

REV NO. 0


GENERAL ELECTRIC
36C764108AA
ELEMENTARY DIAGRAM
SYNCHRONIZER II
STANDARD LINE
193X198AAG01
+20V
TO PIN 8 OF OP AMP 0A376
TO PIN 8 OF OP AMP 0A378
TO PIN 8 OF OP AMP 0A377
TO PIN 8 OF OP AMP 0A375
A COM
TO PIN 4 OF OP AMP 0A376
TO PIN 4 OF OP AMP 0A378
TO PIN 4 OF OP AMP 0A377
TO PIN 4 OF OP AMP 0A375
-20V
+5V
D COM
VFB
FV
R375 330Ω
R376 220Ω
R377 220Ω
R378 220Ω
R379 100Ω
R380 220Ω
R381 220Ω
R382 220Ω
C152 4.7MF
C153 4.7MF
C154 4.7MF
C156 .01MF
C375 .01MF
C376 .01MF
C377 .01MF
C378 .01MF
C379 .01MF
C380 .01MF
C381 .01MF
C382 .01MF
C385 .033MF
R399 47.5K
R400 47K
R401 68.1K
R402 10K
R403 39.2K
R404 22K
R405 39.2K
R406 390K
R407 12K
R408 12K
R409 4.7K
R410 6.8K
R411 100Ω
R431 33K
R432 15K
R433 1K
R434 2.2K
R435 100Ω
R459 47Ω
C386 .22MF
C387 .22MF
C388 .01MF
C389 .22MF
OA375 A58
T389 U1898E
T390 U1898E
T391 2N6027
T379 2N5366
T380 2N4424
D375
D376
D377
D378
FXL
ITWS
OTWX3
R383 1K
R384 3.9K
R385 1K
R386 1K
R387 3.9K
R388 1K
C383 .001MF
C384 .001MF
T375 2N4424
T376 2N4424
D36 IC3
D46 IC2
D53 IC1
D53 IC4
R389 2.2K
R390 10K
R391 10K
R392 10K
R394 2.2K
R395 8.2K
R396 82K
R397 10K
R398 221K
R393 221K
T377 2N4424
T378 2N5366
A COM
D COM
TO SHEET 2
REVISIONS
PRINTS TO
5B(2)22
5D(BK)
5(QC)B
5QQ(BK)5
5(TS)50
5(S)
REV NO. 0
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY
MADE BY J. Martin JULY 12, 1972
SPEED VARIATOR
ERIE, PA.
36C764108AA

GENERAL ELECTRIC

36C764108AA

ELEMENTARY DIAGRAM
SYNCHRONIZER II

STANDARD LINE
193X198AAG01

FROM SHEET 1

D379
C390 .22MF
R412 27K
R420 15K *
R419 150K
C391 .01MF
R416 10K *
R414 10K *
R415 4.7K
R417 6.81K *
R413 100K
OA376 A58
OA377 A58
OA378 A58
T388 U1898E
D380
R418 4.7K
R421 10K
TRIANGLE WAVE 8
+20V
R428 2.2K
R429 15K
R430 100K
C392 .001MF
R427 100K
D381
T383 2N4424
T384 2N4424
+5V
T382 2N5366
R424 100K
D382
T381 2N4424
R422 10K
D386
R423 3.3K
R425 33K
R426 5.6K
-20V
ACOM

CFA 23
R441 4.75K *
R442 18.2K *
R443 4.7K
CSA 22

CFB 20
R444 4.75K *
R445 18.2K *
R446 4.7K
CSB 18

CFC 14
R447 4.75K *
R448 18.2K *
R449 4.7K
CSC 12

R450 22Ω
R451 * 2.21K
R452 * 6.81K
C394 .01MF
R453 * 6.81K
R454 * 10K
T385 2N5366
T386 2N5366
C393 4.7MF
R456 1K *
R457 1K *
R455 * 475Ω
R458 * 332Ω
C395 .1MF
R436 3.3K
R437 2.2K
D383
OFT 9
T387 2N4424
D384
D385
R438 560Ω
R439 560Ω

| | |
|---|---|
| 5B(2)22 | |
| 5D(8K) | |
| 5(QC)8 | |
| 5QC(8K)5 | |
| 5(TS)50 | |
| 5(S) | |

C. Matta JULY 11, 1972

SPEED VARIATOR

36C764108AA

GENERAL ELECTRIC

36C764108AA
CONT ON SHEET FL SH NO 3

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING

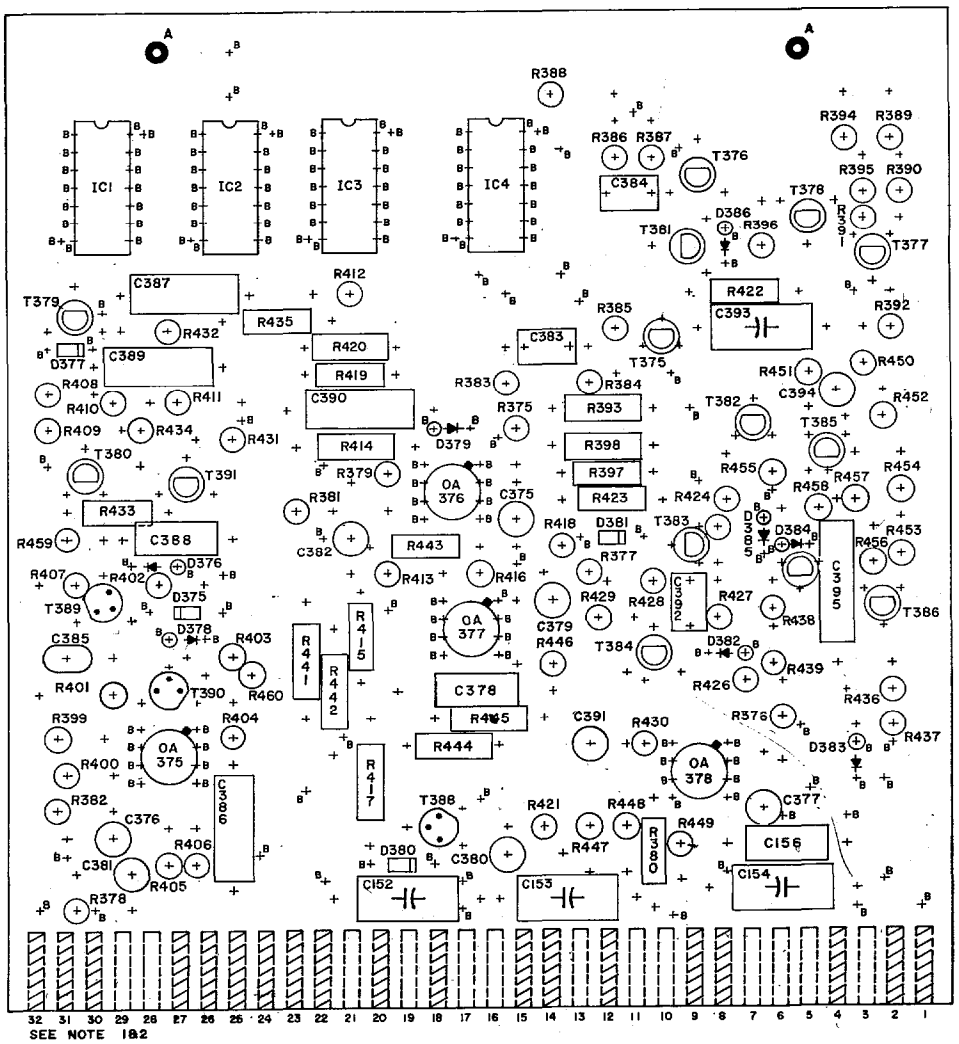
| APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS | DECIMALS | ANGLES |
|---|---|---|---|---|
| 718A515 | √ | ± | ± | ± |

TITLE PRINTED CIRCUIT DIAGRAM
SYNCHRONIZER II
FIRST MADE FOR STANDARD LINE
193X198AAG01

## HOLE TABULATION

ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

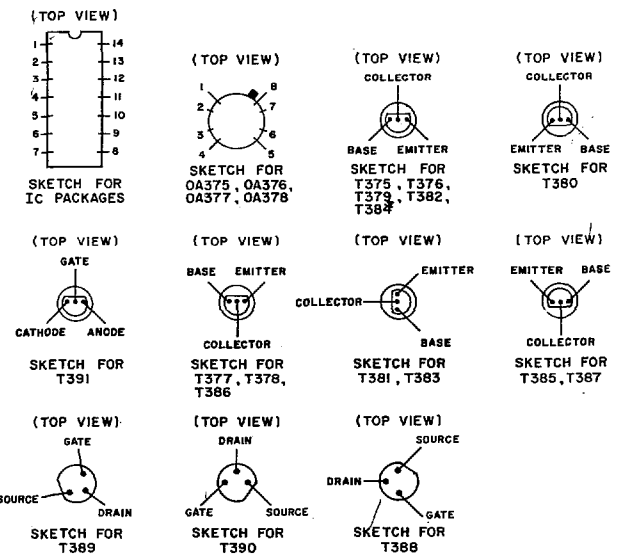
| LOC. | DIA. | QUAN. |
|---|---|---|
| A | .157 | 2 |
| B | .032 | 164 |




### NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$

### TRANSISTOR SKETCHES




| 2X | 36A353762AA005 | 36A353763AA005 | | |
|---|---|---|---|---|
| SCALE | REAR ETCHED BOARD DWG. | FRONT ETCHED BOARD DWG. | FRONT SPACE | BACK UNIT REQUIREMENT |

REVISIONS

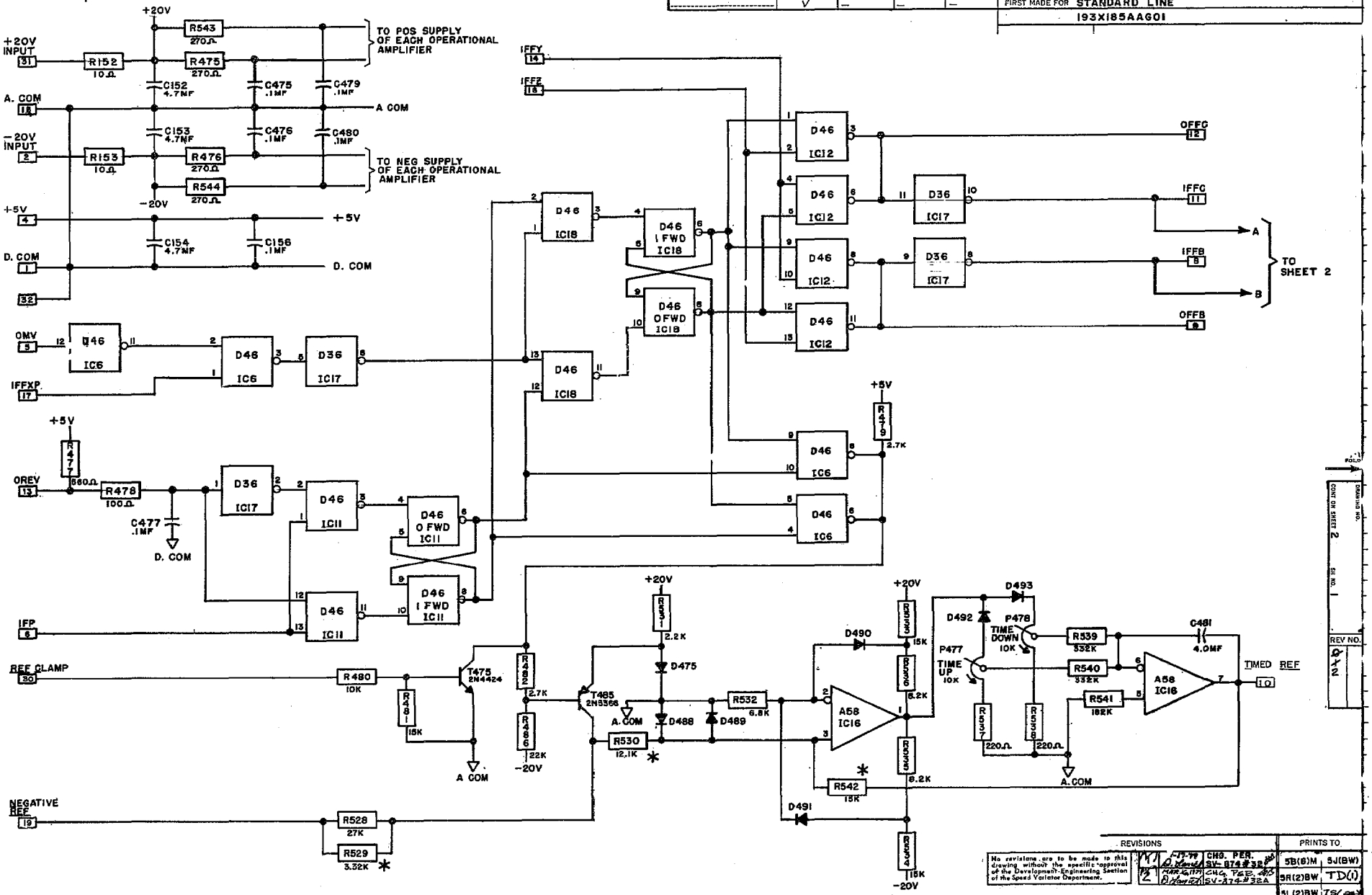
No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

PRINTS TO

| | |
|---|---|
| 5QC(5) | 5B (S) |
| | 5D(BK) |
| | 5B (2) |
| | |
| | 5QC(B) |

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

MADE BY G. Baker OCT. 24, 1972 APPROVED
SPEED VARIATOR ERIE
36C764108AA CONT ON SHEET FL SH NO 3




GENERAL ELECTRIC

36C762855AB
CONT ON SHEET 2 SH NO 1

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

| APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS FRACTIONS | DECIMALS | ANGLES |
|---|---|---|---|---|
| | ✓ | ± | ± | ± |

TITLE ELEMENTARY DIAGRAM
REVERSE LOGIC

FIRST MADE FOR STANDARD LINE
193X185AAG01

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

| REV | | | |
|---|---|---|---|
| M | 1-17-74 D. Monek | CHG. PER. SV-874 #32 | |
| N | 2-15-74 D. Monek | CHG. PER. SV-874 #32A | |

PRINTS TO

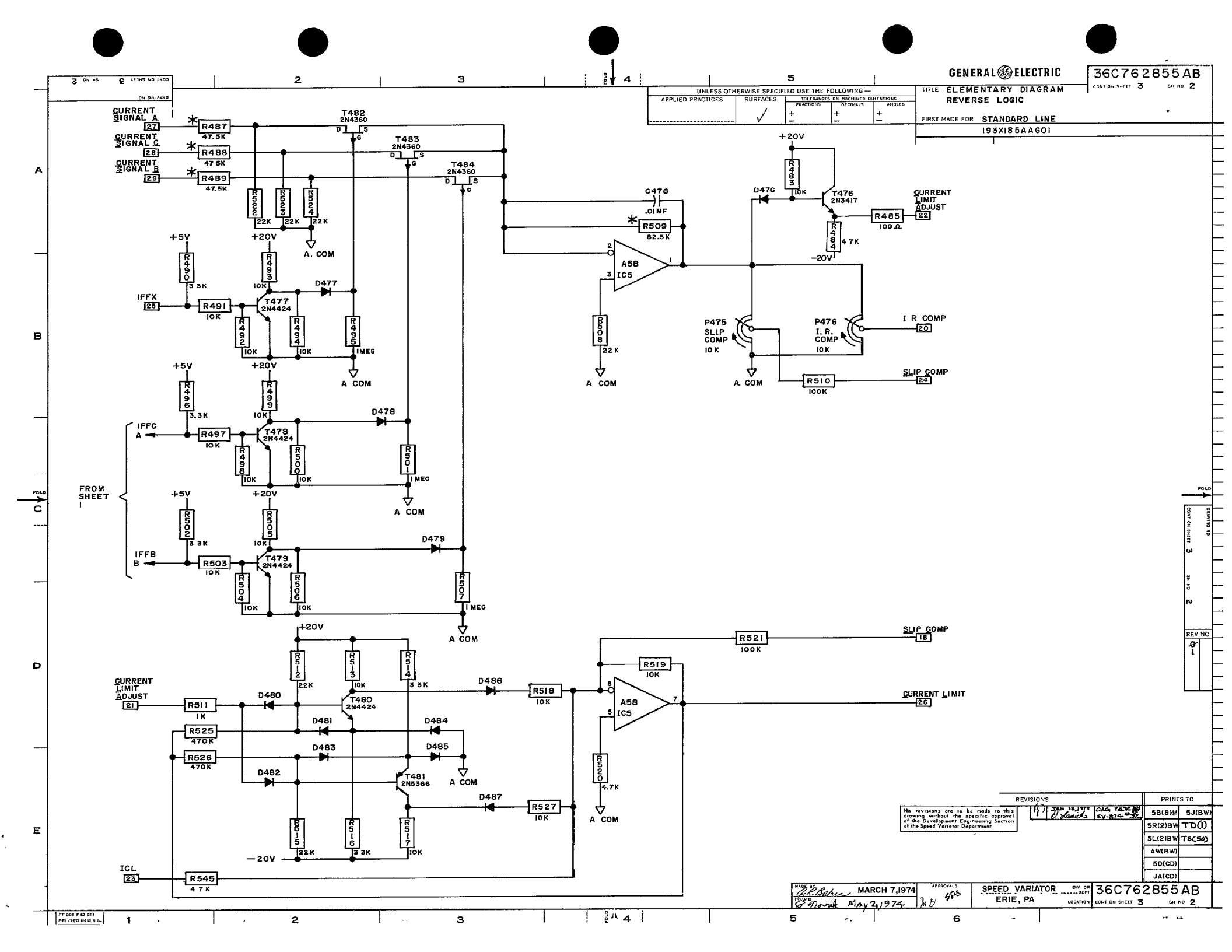
| | |
|---|---|
| 5B(8)M | 5J(BW) |
| 5R(2)BW | TD(I) |
| 5L(2)BW | TS(CD) |
| AW(BW) | |
| 5D(CD) | |
| JA(CD) | |

MADE BY L.P. Baker MARCH 7, 1974
CHECKED D. Monek MAY 2, 1974
APPROVALS M.D. GNC

SPEED VARIATOR DIV OR DEPT
ERIE, PA. LOCATION

36C762855AB
CONT ON SHEET 2 SH NO 1

FF-309-F12-681 PRINTED IN U.S.A.


GENERAL ELECTRIC
36C762855AB
CONT ON SHEET 3 SH NO 2
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES
SURFACES
TOLERANCES ON MACHINED DIMENSIONS
FRACTIONS
DECIMALS
ANGLES
TITLE ELEMENTARY DIAGRAM
REVERSE LOGIC
FIRST MADE FOR STANDARD LINE
193X185AAG01
CURRENT SIGNAL A
27
CURRENT SIGNAL C
28
CURRENT SIGNAL B
29
R487
47.5K
R488
47.5K
R489
47.5K
T482
2N4360
T483
2N4360
T484
2N4360
R522
22K
R523
22K
R524
22K
A. COM
+5V
+20V
R490
3.3K
R493
10K
IFFX
25
R491
10K
T477
2N4424
D477
R492
10K
R494
10K
R495
1MEG
A COM
R496
3.3K
R499
10K
IFFC
A
R497
10K
T478
2N4424
D478
R498
10K
R500
10K
R501
1MEG
FROM SHEET 1
R502
3.3K
R505
10K
IFFB
B
R503
10K
T479
2N4424
D479
R504
10K
R506
10K
R507
1MEG
C478
.01MF
R509
82.5K
A58
IC5
R508
22K
R483
10K
D476
T476
2N3417
CURRENT LIMIT ADJUST
22
R485
100Ω
R484
4.7K
-20V
P475
SLIP COMP
10K
P476
I. R. COMP
10K
I R COMP
20
SLIP COMP
24
R510
100K
R521
100K
SLIP COMP
18
R519
10K
CURRENT LIMIT
26
R512
22K
R513
10K
R514
3.3K
CURRENT LIMIT ADJUST
21
R511
1K
D480
T480
2N4424
D481
D484
R525
470K
D483
D485
R526
470K
D482
T481
2N5366
D486
R518
10K
D487
R527
10K
R515
22K
R516
3.3K
R517
10K
-20V
ICL
23
R545
47K
R520
4.7K
REVISIONS
No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department
PRINTS TO
5B(8)M
5J(BW)
5R(2)BW
TD(1)
5L(2)BW
Ts(50)
AW(BW)
5D(CD)
JA(CD)
MADE BY
MARCH 7,1974
ISSUED
May 2,1974
APPROVALS
SPEED VARIATOR
ERIE, PA
DIV OR DEPT
LOCATION
36C762855AB
CONT ON SHEET 3 SH NO 2

GENERAL ELECTRIC

36C762855AB CONT ON SHEET FL SH NO 3

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING

| APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS FRACTIONS | DECIMALS | ANGLES |
|---|---|---|---|---|
| 716A515 | ✓ | ± | ± | ± |

TITLE PRINTED CIRCUIT DIAGRAM
REVERSE LOGIC

FIRST MADE FOR STANDARD LINE

193X185AAG01

## HOLE TABULATION

ALL HOLES 032 DIA. EXCEPT THE HOLES TABULATED BELOW

| LOC. | DIA. | QUAN |
|---|---|---|
| A | .157 | 2 |
| B | .040 | 213 |
| C | .052 | 2 |

## NOTES

1 INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS

2 CROSS HATCHED TABS INDICATES TABS USED.

3 CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$

## TRANSISTOR LEAD SKETCHES

(TOP VIEW) EMITTER, COLLECTOR, BASE — SKETCH FOR T477, T478 T479, T481

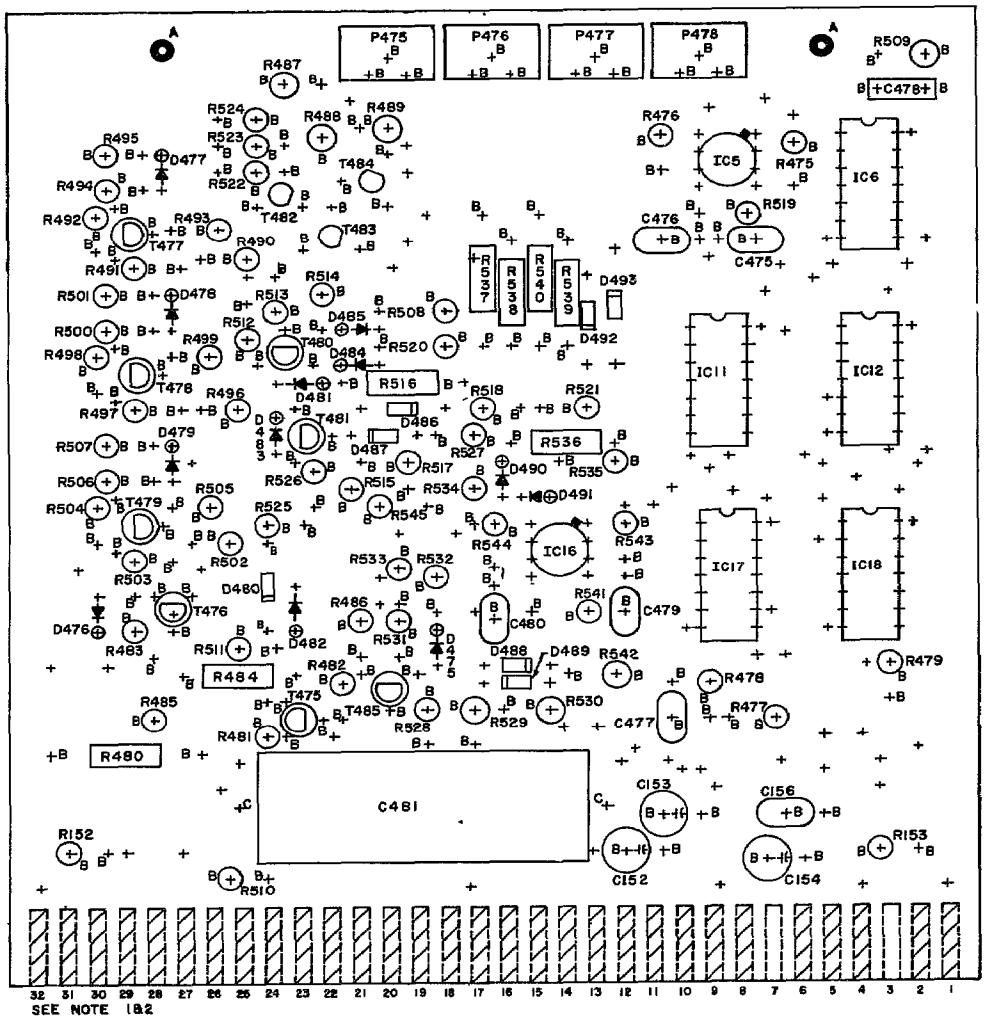
(TOP VIEW) BASE, COLLECTOR, EMITTER — SKETCH FOR T475

(TOP VIEW) BASE, EMITTER, COLLECTOR — SKETCH FOR T476, T485

(TOP VIEW) COLLECTOR, BASE, EMITTER — SKETCH FOR T480

(TOP VIEW) DRAIN, GATE, SOURCE — SKETCH FOR T482, T483

(TOP VIEW) SOURCE, DRAIN, GATE — SKETCH FOR T484




| 2X | 36A352970AB005 | 36A352971AB005 | 2 | |
|---|---|---|---|---|
| SCALE | REAR ETCHED BOARD DWG. | FRONT ETCHED BOARD DWG. | FRONT SPACE | BACK UNIT REQUIREMENT |

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

| | | | |
|---|---|---|---|
| 1 | E.R. Baker July 8, 1974 | F&B | CHG REC # 5V-874 #22 |
| 2 | W. Reed May 16, 1979 | CHG REC # 5V-874 #22 | |
| 3 | D. Novak | | |

PRINTS TO

| | |
|---|---|
| 5B(8)M | 5V(BW) |
| 5R(2)BW | TDC(1) |
| 5L(2)BW | TSC(2) |
| 2AW(BW) | |
| 5D(CD) | |
| 4A(CD) | |

MADE BY E.R. Baker MARCH 8, 1974
ISSUED D. Novak MAY 2, 1974

APPROVALS M.G. F&B

SPEED VARIATOR DIV. OR DEPT
ERIE, PA. LOCATION

36C762855AB CONT ON SHEET FL SH NO 3

SCHEMATIC DIAGRAM

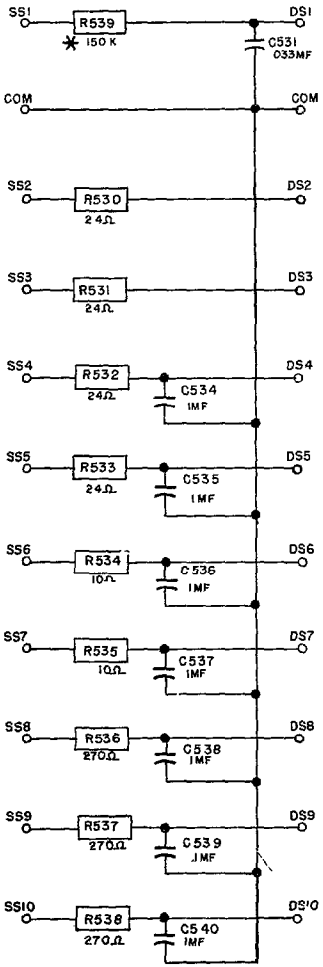
GENERAL ELECTRIC

36C762868AA

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING

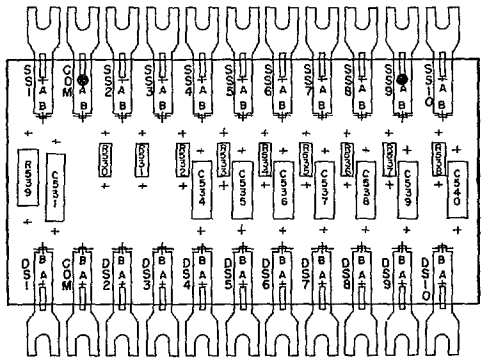
| APPLIED PRACTICES | SURFACES | | |
|---|---|---|---|
| 718A515 | ✓ | | |

PRINTED CIRCUIT DIAGRAM
SIGNAL FILTER
P STANDARD LINE
193X189AAG01




✱ — INDICATES 1% RESISTANCES TOLERANCE

FRONT VIEW SHOWING LOCATION OF COMPONENTS




HOLE TABULATION

ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

| LOC | DIA | QUAN |
|---|---|---|
| A | .157 | 22 |
| B | .101 | 22 |

NOTE. 1 CARD SIZE, $2.85^{+.020}$ X $5.130^{+.002}_{-.008}$

36C762868AA

| SCALE | ETCHED CIRCUIT BOARD DWG | SPACE UNIT REQUIREMENT FRONT | BACK |
|---|---|---|---|
| FULL | 9428522AA004 | | |

REVISIONS

| | | | |
|---|---|---|---|
| 1 | Schmidt JAN 22, 1971 | KWP | 5QC(BK)5 (S) 5D(BK) 5(PCM) 50 5B(2) 5(TS)50 5(QC)B |

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

APPROVALS

T. Schmidt June 2, 1970 EAS

JAN 22, 1971 KWP

SPEED VARIATOR
ERIE, PA.

36C762868AA

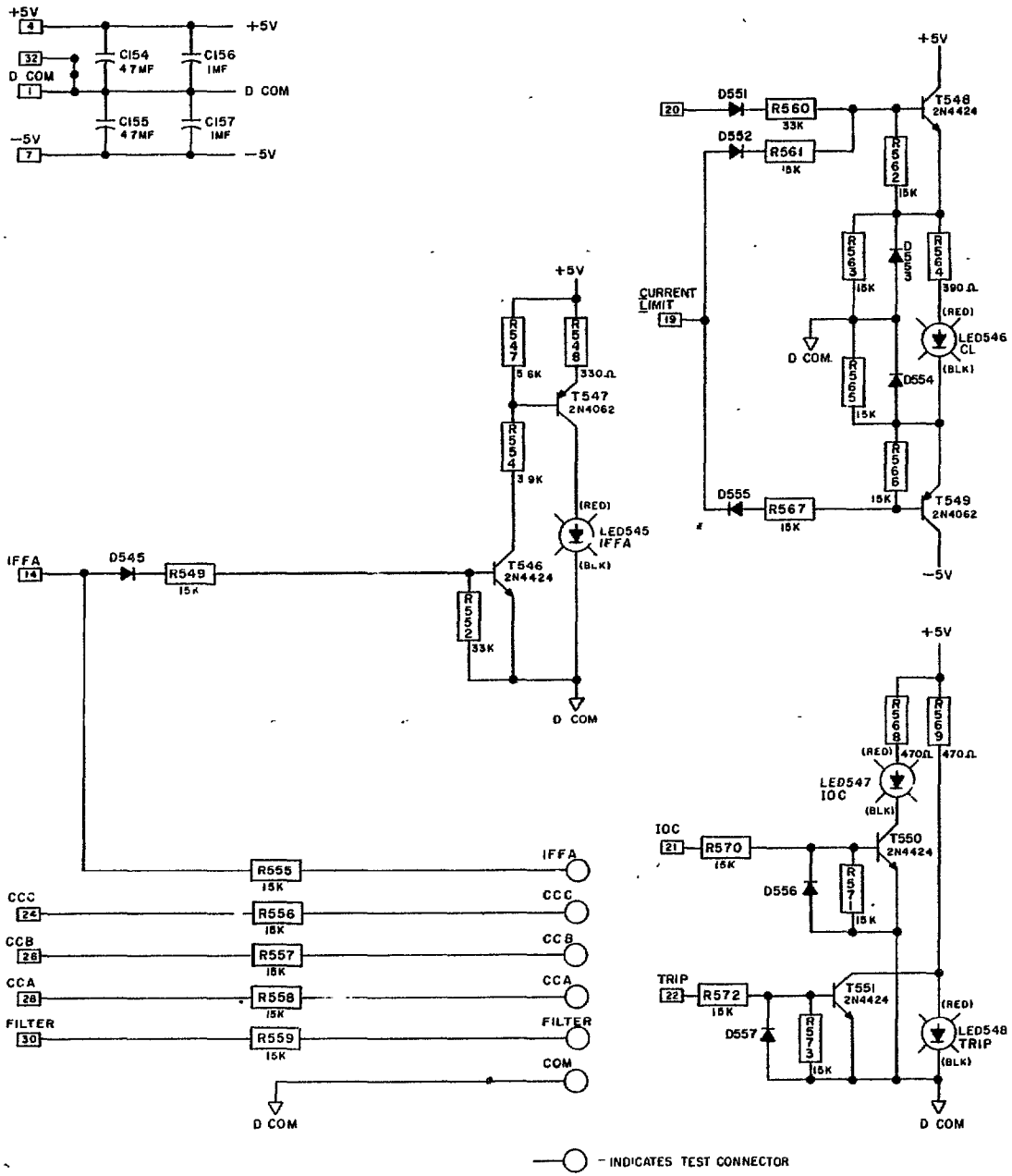
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

| UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING – | | | | |
|---|---|---|---|---|
| APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS | | |
| | | FRACTIONS | DECIMALS | ANGLES |
| 718A515 | ✓ | ± | ± | ± |

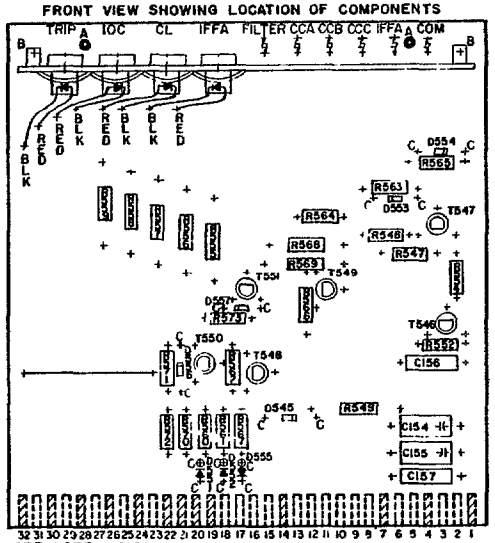
GENERAL ELECTRIC

36C762854AA

TITLE PRINTED CIRCUIT DIAGRAM
MONITOR
FIRST MADE FOR STANDARD LINE
193X184AAG01




FRONT VIEW SHOWING LOCATION OF COMPONENTS




SEE NOTE 1 & 2

HOLE TABULATION

ALL HOLES 040 DIA. EXCEPT THE HOLES TABULATED BELOW

| LOC | DIA | QUAN |
|---|---|---|
| A | .157 | 2 |
| B | .078 | 2 |
| C | 032 | 16 |

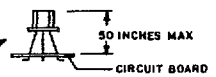
NOTE 1 INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.

2 CROSS HATCHED TABS INDICATE TABS USED

3 CARD SIZE, 5.500 +.000 −.015 X 5.130 +.002 −.008

4 ALL TRANSISTORS SHALL BE MOUNTED TO A .50 INCH MAXIMUM ABOVE THE CARD SURFACE AS SHOWN HERE

(SIDE VIEW)




TRANSISTOR LEAD SKETCHES




36C762854AA
REV 110
0

| SCALE | ETCHED CIRCUIT BOARD DWG. | SPACE UNIT REQUIREMENT | |
|---|---|---|---|
| | | FRONT | BACK |
| FULL | 942B526AA004 | 1 | 0 |

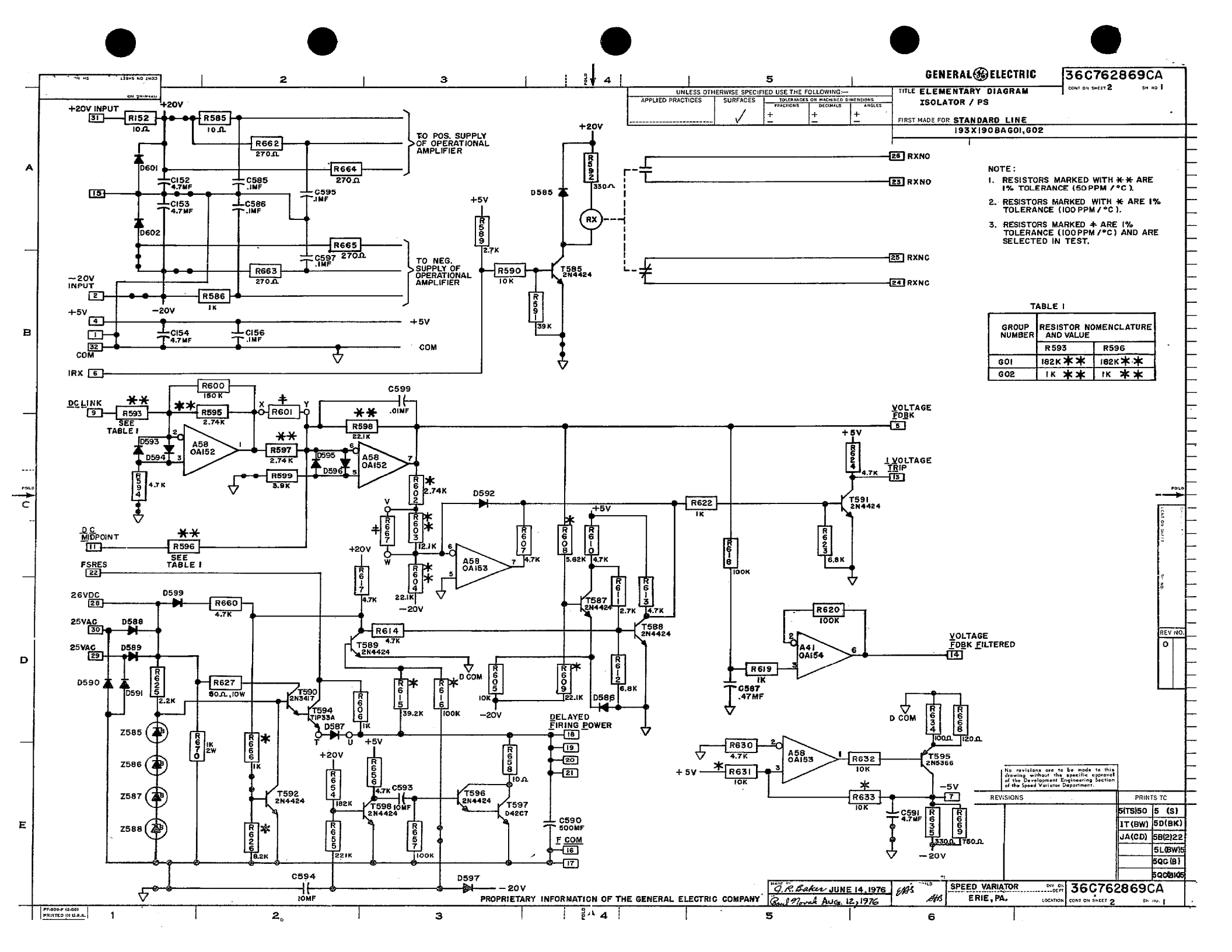
REVISIONS

PRINTS

| | |
|---|---|
| 5QC(BK)5 (S) | |
| 5D(BK) | |
| 5B(2) | |
| 5(TS)50 | |
| 5(QC)B | |

| MADE BY | APPROVALS | | | |
|---|---|---|---|---|
| Q Britton JUNE 3, 1970 | GAS | SPEED VARIATOR | FIN OR DRWG | 36C762854AA |
| ISSUED K Simmons FEB 15 1971 | KRK | ERIE, PENNA | LOCATION | SH OR SHEET |

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY


GENERAL ELECTRIC
36C762869CA
TITLE ELEMENTARY DIAGRAM
ISOLATOR / PS
FIRST MADE FOR STANDARD LINE
193X190BAG01,G02
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES
SURFACES
TOLERANCES ON MACHINED DIMENSIONS
FRACTIONS
DECIMALS
ANGLES
+20V INPUT
−20V INPUT
+5V INPUT
COM
IRX
TO POS. SUPPLY OF OPERATIONAL AMPLIFIER
TO NEG. SUPPLY OF OPERATIONAL AMPLIFIER
R152 10Ω
R585 10Ω
R662 270Ω
R664 270Ω
R665 270Ω
R663 270Ω
R586 1K
C152 4.7MF
C153 4.7MF
C154 4.7MF
C585 .1MF
C586 .1MF
C595 .1MF
C597 .1MF
C156 .1MF
D601
D602
R589 2.7K
R590 10K
R591 39K
T585 2N4424
D585
R592 330Ω
RX
RXNO
RXNC
NOTE:
1. RESISTORS MARKED WITH ** ARE 1% TOLERANCE (50 PPM / °C).
2. RESISTORS MARKED WITH * ARE 1% TOLERANCE (100 PPM / °C).
3. RESISTORS MARKED # ARE 1% TOLERANCE (100 PPM / °C) AND ARE SELECTED IN TEST.
TABLE I
GROUP NUMBER
RESISTOR NOMENCLATURE AND VALUE
R593
R596
G01 182K** 182K**
G02 1K** 1K**
DC LINK
SEE TABLE I
R600 150K
R595 2.74K
R601
R593
R594 4.7K
D593
D594
A58 OA152
R597 2.74K
R599 3.9K
D595
D596
R598 22.1K
C599 .01MF
R602 2.74K
VOLTAGE FDBK
+5V
R624 4.7K
I VOLTAGE TRIP
T591 2N4424
R622 1K
R623 6.8K
R618 100K
DC MIDPOINT
R596
FSRES
R603 12.1K
R604 22.1K
R607 4.7K
A58 OA153
D592
R608 5.62K
R610 4.7K
R611 2.7K
R613 4.7K
T587 2N4424
T588 2N4424
R614 4.7K
T589 2N4424
R617 4.7K
R620 100K
A41 OA154
VOLTAGE FDBK FILTERED
R619 1K
C587 .47MF
26VDC
25VAC
D599
D588
D589
D590
D591
R660 4.7K
R625 2.2K
R627 80Ω, 10W
T590 2N3417
T594 TIP33A
D587
R605 10K
R609 22.1K
R612 6.8K
D586
D COM
DELAYED FIRING POWER
R615 39.2K
R616 100K
R606 1K
R670 1K 2W
Z585
Z586
Z587
Z588
R666 1K
R626 8.2K
T592 2N4424
R654 182K
R656 4.7K
C593 10MF
T598 2N4424
R655 221K
R657 100K
T596 2N4424
T597 D42C7
R658 10Ω
C590 500MF
F COM
C594 10MF
D597
−20V
R630 4.7K
R631 10K
R632 10K
R633 10K
A58 OA153
T595 2N5366
R634 100Ω
R668 120Ω
R635 330Ω
R669 750Ω
C591 4.7MF
No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.
REVISIONS
PRINTS TO
5(TS)50 5 (S)
1T(BW) 5D(BK)
JA(CD) 5B(2)22
5L(BW)5
5QC(B)
5QC(B)05
REV NO. 0
CONT ON SHEET 2
SH NO 1
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY
MADE BY G.R. Baker JUNE 14, 1976
APPROVALS Paul Novak Aug. 12, 1976
SPEED VARIATOR
ERIE, PA.
DIV. OR DEPT
LOCATION
36C762869CA
CONT ON SHEET 2
SH NO 1
PF-600-F (2-68) PRINTED IN U.S.A.

GENERAL ELECTRIC

36C762869CA
CONT ON SHEET FL SH NO 2

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING -

| APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS FRACTIONS | DECIMALS | ANGLES |
|---|---|---|---|---|
| 718A515 | ✓ | ± | ± | ± |

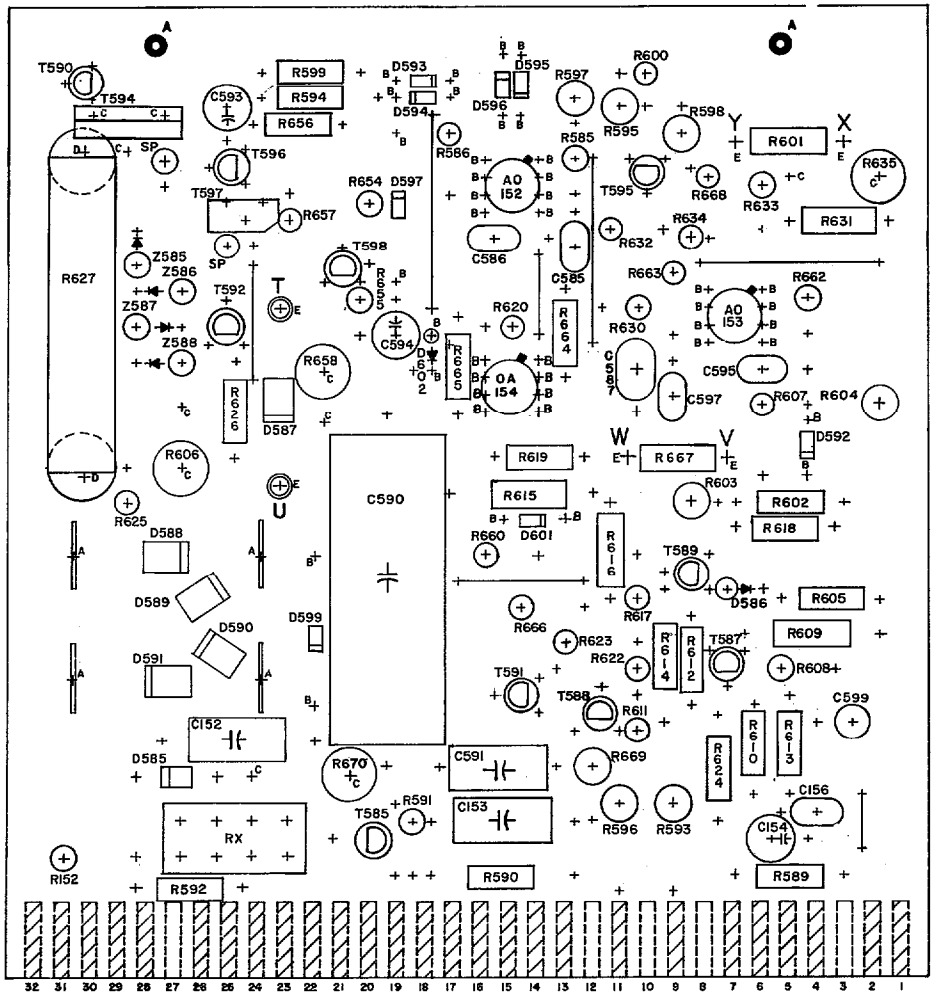
TITLE PRINTED CIRCUIT DIAGRAM
ISOLATOR / PS

FIRST MADE FOR STANDARD LINE
193X190BA601, 602

## HOLE TABULATION

ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

| LOC. | DIA. | QUAN. |
|---|---|---|
| A | .157 | 6 |
| B | .032 | 42 |
| C | .062 | 11 |
| D | .052 | 2 |
| E | .078 | 6 |



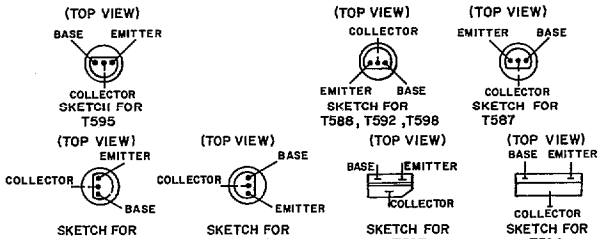

32 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1

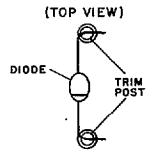
SEE NOTE 1&2

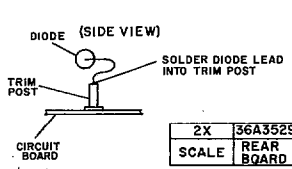
## NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.015} \times 5.130^{+.002}_{-.008}$

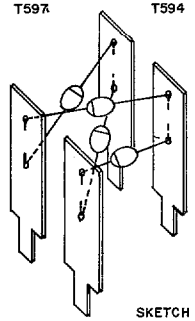
## TRANSISTOR LEAD SKETCHES




SKETCH FOR MTG D587



SKETCH FOR DIODE ASM. (D588, D589, D590, D591)

| 2X | 36A352969CA004 | | 2 | |
|---|---|---|---|---|
| SCALE | REAR ETCHED BOARD DWG. | FRONT ETCHED BOARD DWG. | FRONT SPACE UNIT REQUIREMENT | BACK |

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

PRINTS TO

| | |
|---|---|
| 5QC(B) 5 | (S) |
| 5QC(BK) 5 | 5D(BK) |
| | 5B(2)22 |
| | 5TS(SC) |

REV NO. 0

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

| MADE BY | DATE | APPROVALS | | DIV OR DEPT | |
|---|---|---|---|---|---|
| J.R. Baker | MAR. 13, 1975 | | | SPEED VARIATOR | 36C762869CA |
| ISSUED Gerald Novak | Aug. 12, 1976 | | | LOCATION ERIE | CONT ON SHEET FL SH NO 2 |

GENERAL ELECTRIC

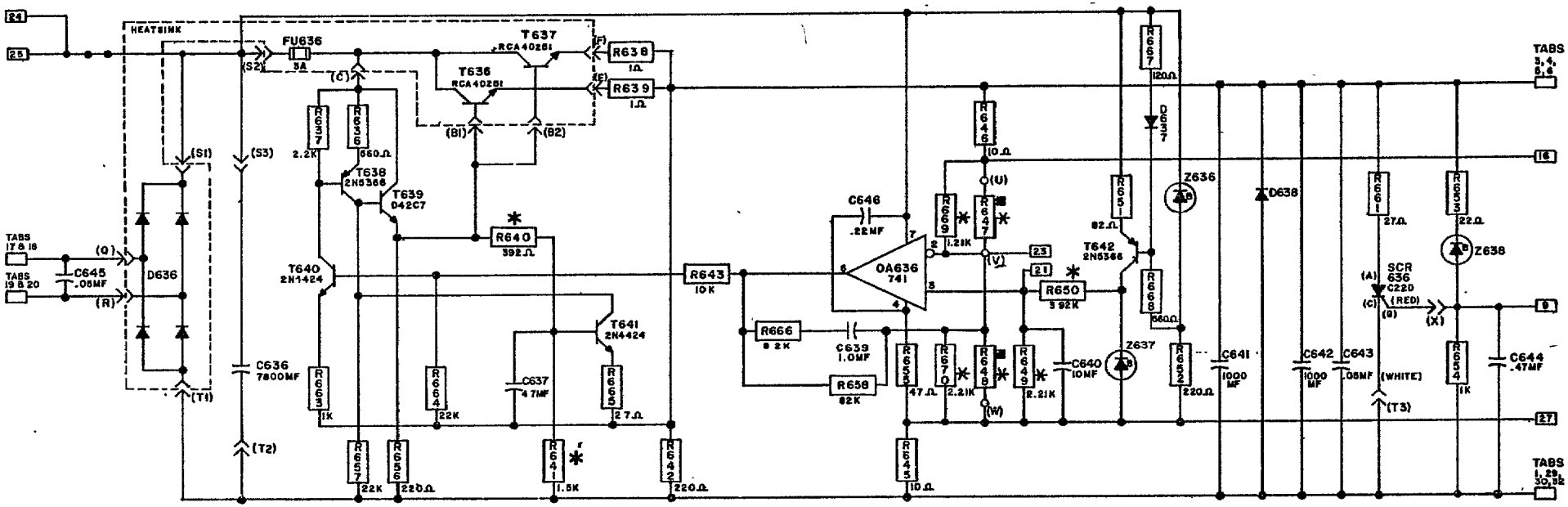
36C762870AB

CONT ON SHEET 2 SH NO. 1

| UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING — | | | | |
|---|---|---|---|---|
| APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS | DECIMALS | ANGLES |
| | ✓ | ± | ± | ± |

TITLE ELEMENTARY DIAGRAM
+5 VOLT POWER SUPPLY

FIRST MADE FOR STANDARD LINE
193X191AB601




✱ INDICATE 1% RESISTOR

■ RESISTORS SELECTED IN TEST

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

| PRINTS TO | |
|---|---|
| 5QC(BK) | 5(S) |
| 5T(BW) | 5D(BK) |
| 5JA(CD) | 5B(2)22 |
| 5AE(BW) | 5L(BW)5 |
| | 5(T5)50 |
| | 5(QC)8 |

DRAWN G. Mann AUG. 25, 1973

APPROVALS

ENGR B. LaRue FEB. 1, 1974

SPEED VARIATOR DIV OR DEPT

ERIE, PA. LOCATION

36C762870AB

CONT ON SHEET 2 SH NO. 1

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV NO 0

GENERAL ELECTRIC

36C762870AB
CONT ON SHEET FL. SH NO 2

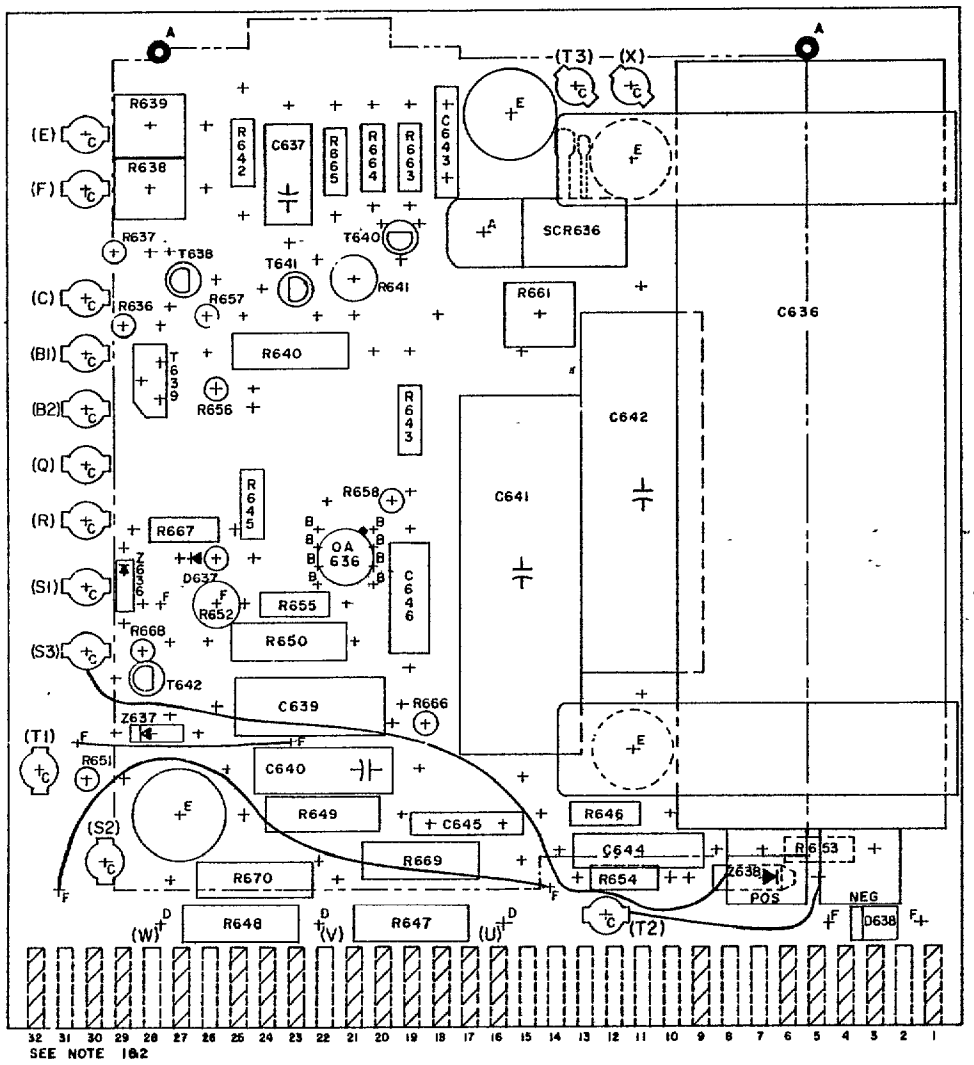
| UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING | | | | |
|---|---|---|---|---|
| APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS | DECIMALS | ANGLES |
| 718A515 | √ | ± | ± | ± |

TITLE PRINTED CIRCUIT DIAGRAM
+ 5 VOLT POWER SUPPLY
FIRST MADE FOR STANDARD LINE
193X191AB G01

## HOLE TABULATION

ALL HOLES 040 DIA EXCEPT THE HOLES TABULATED BELOW

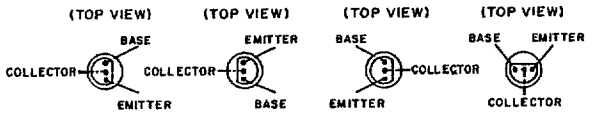
| LOC. | DIA. | QUAN |
|---|---|---|
| A | .157 | 3 |
| B | 032 | 8 |
| C | 101 | 14 |
| D | .078 | 3 |
| E | 187 | 4 |
| F | .062 | 8 |

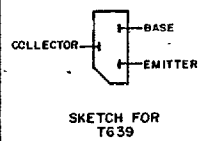



## NOTES

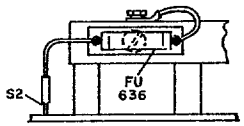
1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$

## TRANSISTOR LEAD SKETCHES

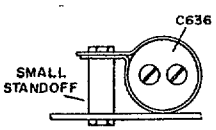



FUSE HOLDER CONNECTION SKETCH
(SIDE VIEW OF HEATSINK FROM TAB END OF CARD)




CAPACITOR MOUNTING SKETCH




## LEAD CONNECTIONS

CONNECT RING TERMINAL OF LEAD S3 TO POSITIVE TERMINAL OF C636
CONNECT RING TERMINAL OF LEAD T2 TO NEGATIVE TERMINAL OF C636
COUPLE RED LEAD FEMALE CONNECTOR TO MALE CONNECTOR X AND SCR636 WHITE LEAD FEMALE CONNECTOR TO MALE CONNECTOR T3
COUPLE REMAINING LEADS WITH FEMALE CONNECTORS TO MALE CONNECTORS BEARING NOMENCLATURE IDENTICAL TO THE LEAD MARKING

| REV NO. |
|---|
| 0 |

| 2X | 36A352968A8004 | 3 | 0 |
|---|---|---|---|
| SCALE | ETCHED CIRCUIT BOARD DWG. | FRONT SPACE UNIT REQUIREMENT | BACK |

REVISIONS

| PRINTS TO | |
|---|---|
| 5L(BW)5 | 5 (S) |
| IT(BW) | 5D(BK) |
| JA(CD) | 5B(2)22 |
| 5AE(BW) | 5T(BK) |
| | 5QC(BK) |
| | 5QC(B) |

MADE BY D. Mattis SEPT 4, 1973
ISSUED B. LaRue FEB. 1, 1974
APPROVALS
SPEED VARIATOR
ERIE, PA
36C762870AB
CONT ON SHEET FL SH NO 2

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

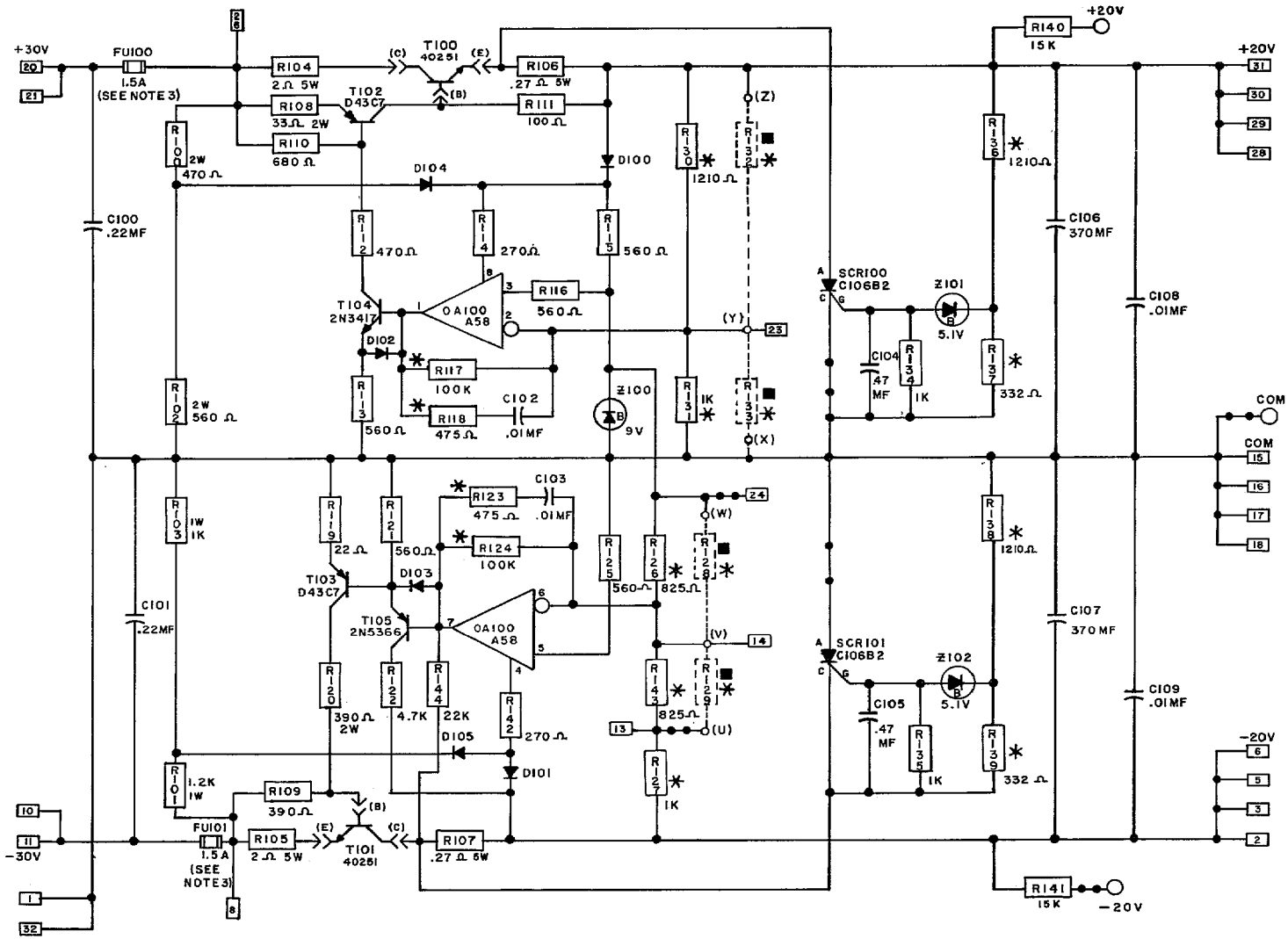
GENERAL ELECTRIC

36C762896AB
CONT ON SHEET 2 SH NO. 1

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

| APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS FRACTIONS | DECIMALS | ANGLES |
|---|---|---|---|---|
| 718A505 | ✓ | ± | ± | ± |

TITLE ELEMENTARY DIAGRAM
20 VOLT POWER SUPPLY CARD
FIRST MADE FOR STANDARD LINE
193X257AAG01




NOTES 1. ✱ — 1% TEMPERATURE COMPENSATED RESISTOR.
■ — RESISTOR SELECTION IN TEST.

2. NUMBERS INSIDE SMALL RECTANGLES INDICATE TAB NUMBERS WHICH CORRESPOND TO MATCHING RECEPTACLE NUMBERS.

3. FUSES ARE INSTRUMENT TYPE, LITTLE FUSE 361001, OR EQUIVALENT.

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

REVISIONS

| PRINTS TO | |
|---|---|
| JA(CD) | 5(S) |
| IT(BW) | 5QC(BK) 5 |
| 5L(BW)5 | 5D(BK) |
| 5AE(BW) | 5B(2)22 |
| | 5(TS)50 |
| | 5(QC)BK |

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

MADE BY G.B.Baker MARCH 15, 1974
APPROVALS
DEPT. E. Novak SEPT. 4, 1974
DEPT. dcg 8/6/74
speed variator DIV. OR DEPT.
ERIE, PA. LOCATION

36C762896AB
CONT ON SHEET 2 SH NO. 1

GENERAL ELECTRIC

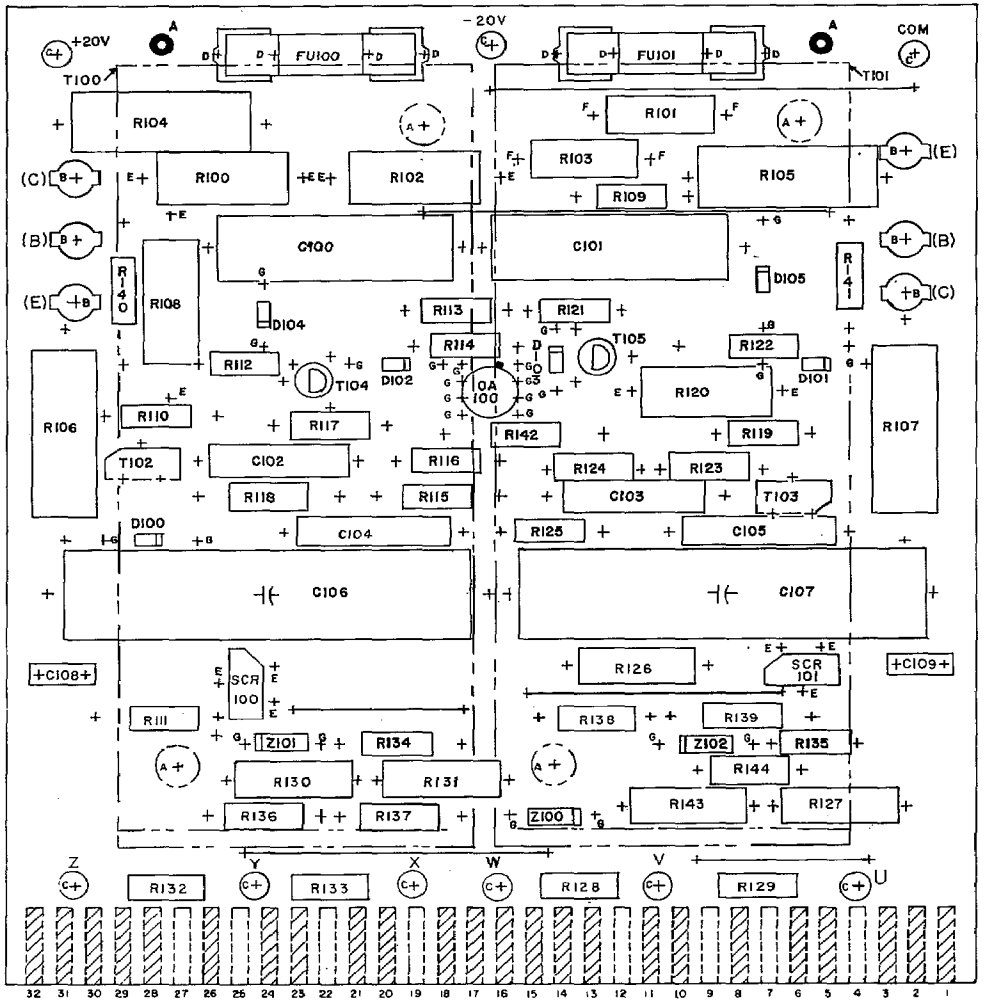
36C762896AB
FL 2

PRINTED CIRCUIT DIAGRAM
20 VOLT POWER SUPPLY

FIRST MADE FOR STANDARD LINE
193X257AA601

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES

| APPLIED PRACTICES | SURFACES | TOLERANCES IN MINUTES |
|---|---|---|
| 718A515 | √ | |




| GROUP | KEY LOCATIONS | | | |
|---|---|---|---|---|
| 01 | 6-7 | 12-13 | 26-27 | — |
| | | | | |
| | | | | |
| | | | | |

HOLE TABULATION

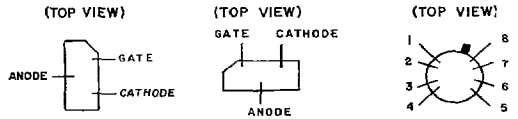
ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

| LOC. | DIA. | QUAN. |
|---|---|---|
| A | .157 | 6 |
| B | .101 | 6 |
| C | .081 | 9 |
| D | .078 | 8 |
| E | .062 | 14 |
| F | .052 | 4 |
| G | .032 | 26 |

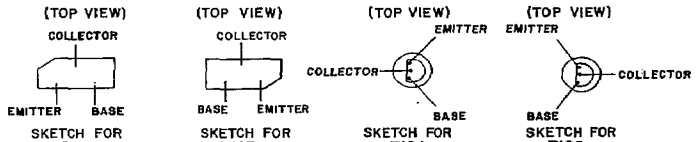
NOTES

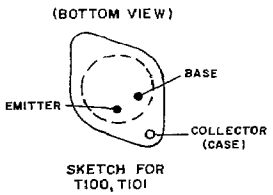
1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$

SCR LEAD SKETCHES




TRANSISTOR LEAD SKETCHES




| | | | | | |
|---|---|---|---|---|---|
| 2X | 36A353740AB004 | | 2 | | |
| SCALE | REAR ETCHED BOARD DWG. | FRONT ETCHED BOARD DWG. | FRONT SPACE UNIT REQUIREMENT | BACK | |

REVISIONS

| PRIV. S.I. | |
|---|---|
| JA(CD) | 5 (S) |
| IT(BW) | 50(BK) |
| 5L(BW) | 55B(2)22 |
| 5AE(BW) | 5QC(BK)5 |
| | 5(TS)50 |
| | 5QC(B) |

MADE BY G. A. Weber MARCH 15, 1974
APPROVED [illegible] speed variator
ISSUED G. Novak SEPT. 4, 1974
ERIE, PA.

36C762896AB
FL 2

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

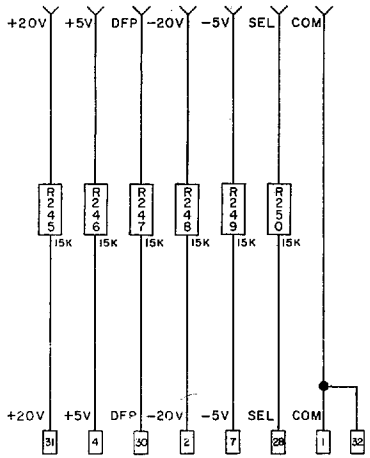
GENERAL ELECTRIC | 36C762858AA

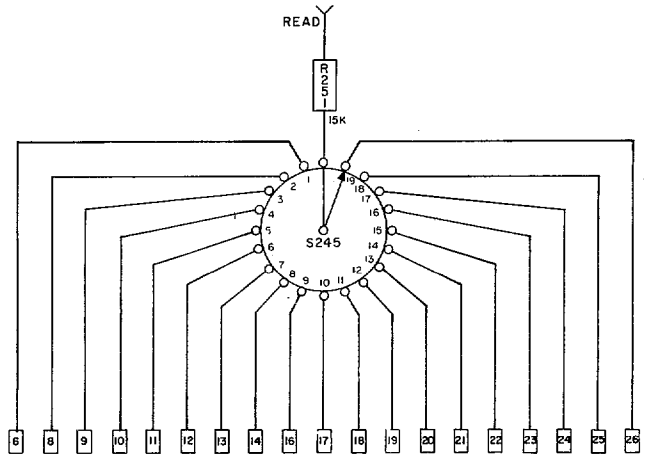
SCHEMATIC DIAGRAM

| UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:- | | | | |
|---|---|---|---|---|
| APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS | | |
| | | FRACTIONS | DECIMALS | ANGLES |
| 718A515 | ✓ | ± | ± | ± |

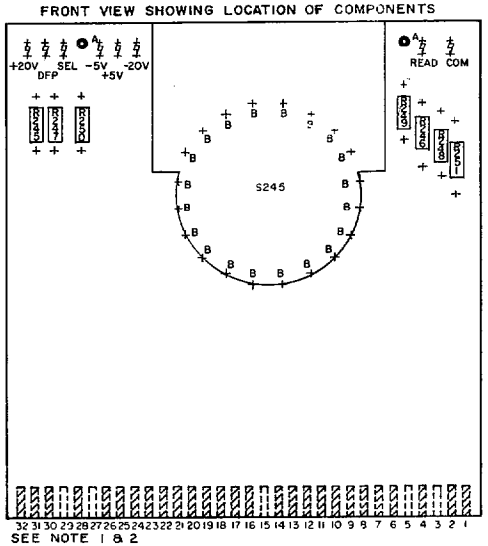
TITLE PRINTED CIRCUIT DIAGRAM
TEST CARD

FIRST MADE FOR STANDARD LINE
193X188AAG01




FRONT VIEW SHOWING LOCATION OF COMPONENTS




SEE NOTE 1 & 2

HOLE TABULATION

ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

| LOC. | DIA. | QUAN. |
|---|---|---|
| A | .157 | 2 |
| B | .062 | 20 |

NOTE 1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATE TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$

36C762858AA REV 0

| SCALE | ETCHED CIRCUIT BOARD DWG. | SPACE UNIT REQUIREMENT | |
|---|---|---|---|
| | | FRONT | BACK |
| FULL | 942B549AA004 | 2 | 0 |

REVISIONS

PRINTS TO
5OC(BK5) 5B(S)
5D(BK
5B(2
5(TS)5(
5(QC)

MADE BY G.R. Baker NOV. 24, 1970 APPROVALS EAS KWP speed varidor ERIE, PA. 36C762858AA

ISSUED | DIV OR DEPT | LOCATION | CONT ON SHEET | SH NO.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC

36C762895AB CONT ON SHEET 2 SH NO. 1

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE ELEMENTARY DIAGRAM
DIVIDER CARD

FIRST MADE FOR STANDARD LINE
193X195AB G01, G02

VFB 12
FXL 10
MV 11
FV 13

R770 12.1K * R771 12.1K * R772 56.2K *
R773 16.2K * R774 16.2K * R775 56.2K *
R776 10K * R777 10K * R778 22.1K *
R779 10K * R780 10K * R781 15K *
R782 33.2K * R783 33.2K * R784 39.2K *

A58 0A751 A58 0A755 A58 0A754 A58 0A753 A58 0A752 A58 0A751

R794 10K R793 150K
R792 2K R791 220K
R790 2K R789 120K
R788 2K R787 120K
R786 3.9K R785 390K

D756 D757 D758 D759 D760 D761

ACOM

R796 82.5K * R797 162K * R798 18K
R795 680K R801 68.1K * R802 162K * R803 22K
R811 470K R806 47.5K * R807 162K * R808 22K
R812 33.2K * R813 162K * R814 22K
R817 * SEE TABLE I R818 221K * R819 27K

D762 D764 D766 D768 D770

R799 1 MEG R804 1 MEG R809 1 MEG R815 1 MEG R820 1 MEG

R800 2K R805 2K R810 2K R816 3.9K R821 2K

D763 D765 D767 D769 D771

−20V

D36 IC11 D46 IC10 D36 IC5 D62 IC6
D36 IC11 D46 IC3 D36 IC5 D62 IC6
D36 IC5 D62 IC4 D46 IC3
D36 IC5 D62 IC4
D36 IC5 D62 IC4 D46 IC3

1 TW SYNC 8
OTWX3 9
IFRTW 8
HFRTW 14

+20V +5V

R2023 681K R2027 100K R2025 10K R2035 4.7K
R2024 15K R2026 470 Ω
T755 2N6027 T752 2N4424
C756 .0015MF

R828 10K *

A B C D E F

FROM SH. 2

TABLE I

| GROUP NO. | RESISTOR NO. AND VALUE |
|---|---|
| | R817 |
| G01 | 10K |
| G02 | 5.62K |

REVISIONS

| | | | |
|---|---|---|---|
| 1 | MOD 1 NOV. 28, 1978 | BAB | |

PRINTS TO

| | |
|---|---|
| 5B(6BW) | 5J(BW) |
| 5R(2)BW | TS(SO) |
| 5L(2)BW | |
| AW(BW) | |
| 5D(CD) | |
| JA(CD) | |

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

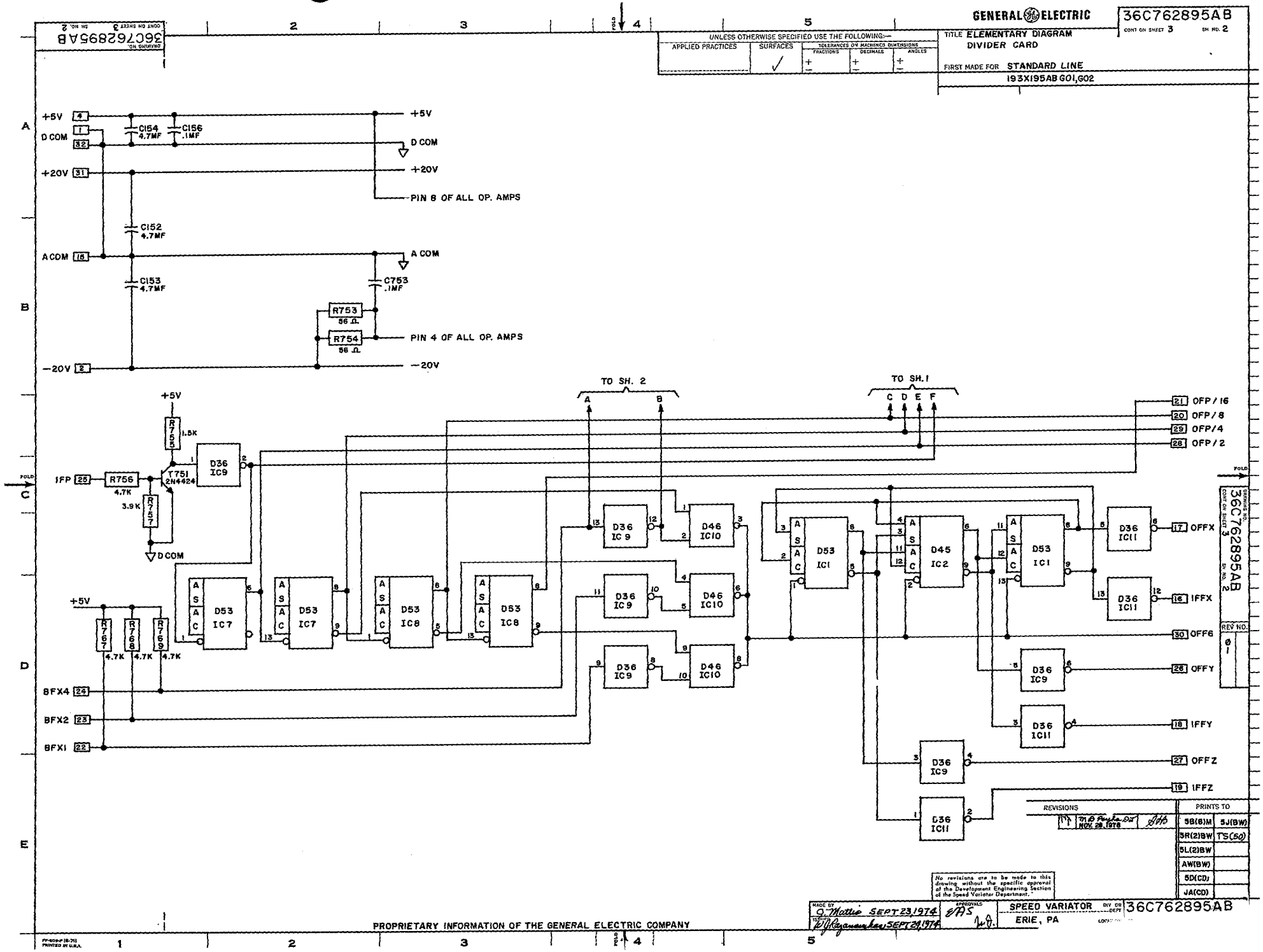
MADE BY O. Matau SEPT 23, 1974
DESIGNED W. Rajagopalan SEPT 24, 1974

APPROVALS BAB M.D.

SPEED VARIATOR DIV OR DEPT
ERIE, PA. LOCATION

36C762895AB CONT ON SHEET 2 SH NO. 1

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY


GENERAL ELECTRIC
36C762895AB
CONT ON SHEET 3
SH NO. 2
TITLE ELEMENTARY DIAGRAM
DIVIDER CARD
FIRST MADE FOR STANDARD LINE
193X195AB G01, G02
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES
SURFACES
TOLERANCES ON MACHINED DIMENSIONS
FRACTIONS
DECIMALS
ANGLES
+5V
D COM
+20V
PIN 8 OF ALL OP. AMPS
C154 4.7MF
C156 .1MF
C152 4.7MF
C153 4.7MF
A COM
C753 .1MF
R753 56Ω
R754 56Ω
PIN 4 OF ALL OP. AMPS
−20V
TO SH. 2
TO SH. 1
IFP
R756 4.7K
R757 3.9K
R755 1.5K
T751 2N4424
D36 IC9
D53 IC7
D53 IC8
D46 IC10
D53 IC1
D45 IC2
D36 IC11
R767 4.7K
R768 4.7K
R769 4.7K
BFX4
BFX2
BFX1
OFP/16
OFP/8
OFP/4
OFP/2
OFFX
IFFX
OFF6
OFFY
IFFY
OFFZ
IFFZ
REVISIONS
PRINTS TO
5B(6)M
5J(BW)
5R(2)BW
TS(50)
5L(2)BW
AW(BW)
5D(CD)
JA(CD)
No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.
MADE BY
APPROVALS
SPEED VARIATOR
ERIE, PA
36C762895AB
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC

36C762895AB CONT ON SHEET FL SH NO 3

| UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING - | | | | |
|---|---|---|---|---|
| APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS | | |
| | | FRACTIONS | DECIMALS | ANGLES |
| 718A515 | ✓ | ± | ± | ± |

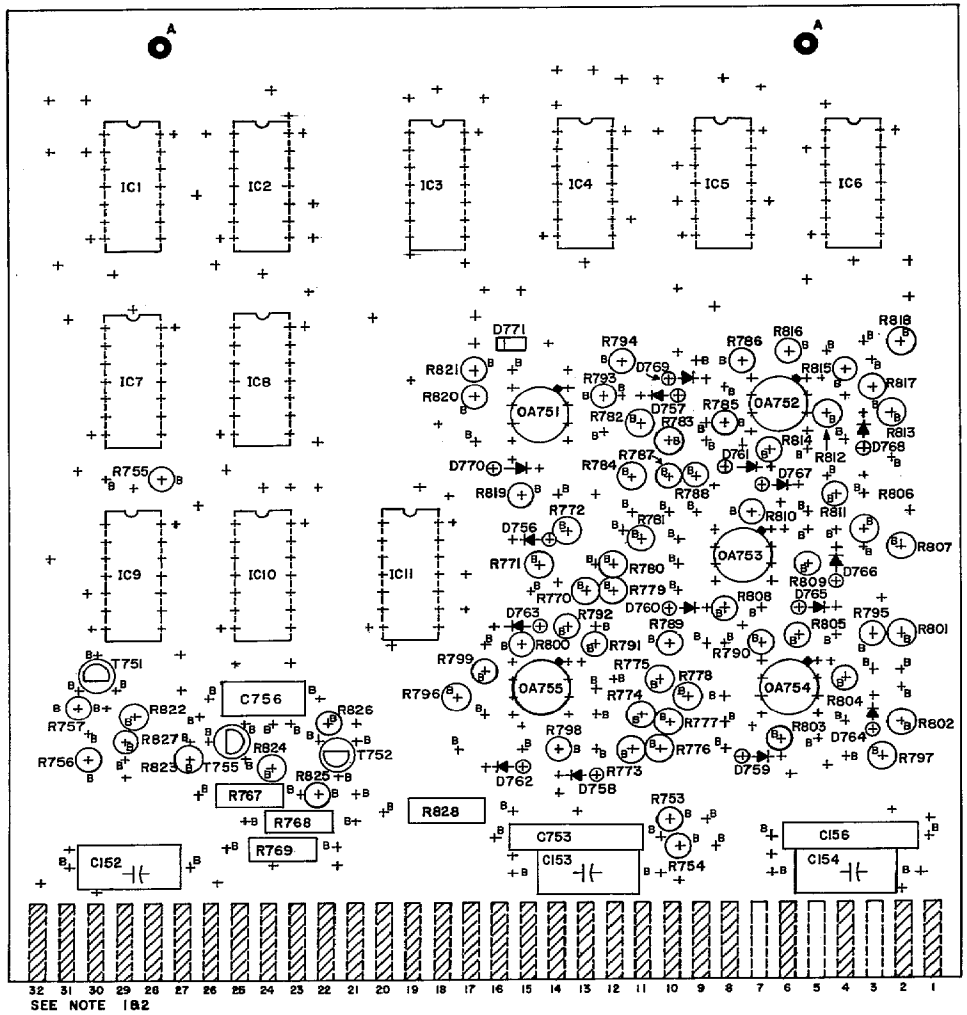
TITLE PRINTED CIRCUIT DIAGRAM
DIVIDER CARD

FIRST MADE FOR STANDARD LINE
193X195AB G01, G02

## HOLE TABULATION

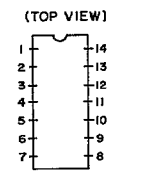
ALL HOLES .032 DIA. EXCEPT THE HOLES TABULATED BELOW

| LOC. | DIA. | QUAN. |
|---|---|---|
| A | .157 | 2 |
| B | .040 | 155 |

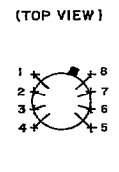



### NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.016}$ X $5.130^{+.002}_{-.008}$




SKETCH FOR IC PACKAGES

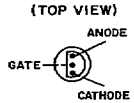



SKETCH FOR 0A751, 0A752, 0A753, 0A754, 0A755

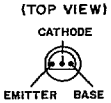
TRANSISTOR LEAD SKETCHES




SKETCH FOR T752




SKETCH FOR T755




SKETCH FOR T751

| 2X | 36A353738AA005 | 36A353739AA005 | 1 | 0 |
|---|---|---|---|---|
| SCALE | REAR ETCHED BOARD DWG. | FRONT ETCHED BOARD DWG. | FRONT SPACE UNIT REQUIREMENT | BACK |

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

M B Boyle 105 NOV. 29, 1978 HB

PRINTS TO

| | |
|---|---|
| 5B(8)M | 5J(BW) |
| 5R(2)BW | T5(50) |
| 5L(2)BW | |
| AW(BW) | |
| 5D(CD) | |
| JA(CD) | |

MADE BY D Morris SEPT 23, 1974
ISSUED W Yaganuchi SEPT 24, 1974
APPROVALS EHS MB

SPEED VARIATOR — DIV OR DEPT
ERIE — LOCATION

36C762895AB CONT ON SHEET FL SH NO 3

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC

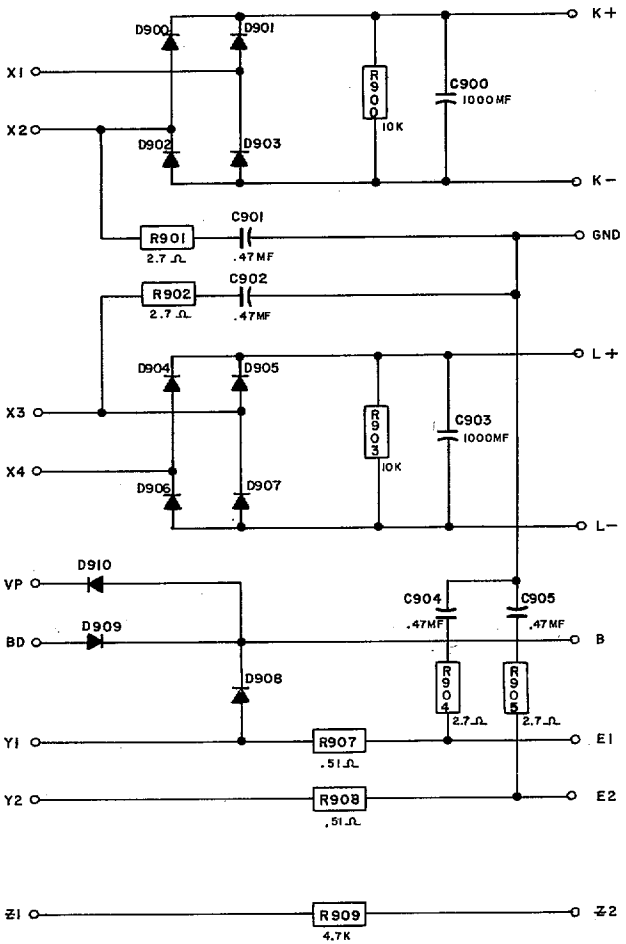
36C764129AA

CONT ON SHEET | SH NO.

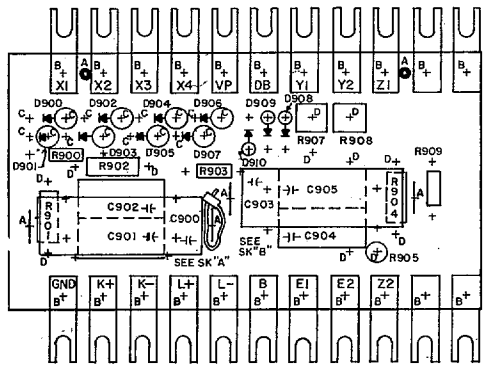
SCHEMATIC DIAGRAM

| UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:– | | | | |
|---|---|---|---|---|
| APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS | | |
| | | FRACTIONS | DECIMALS | ANGLES |
| 718AS15 | ✓ | ± | ± | ± |

TITLE PRINTED CIRCUIT DIAGRAM
POWER INTERFACE
FIRST MADE FOR STANDARD LINE
193X370AA G01




FRONT VIEW SHOWING LOCATION OF COMPONENTS

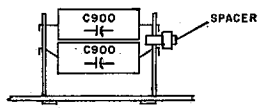



HOLE TABULATION

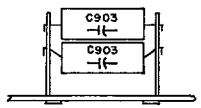
ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

| LOC. | DIA. | QUAN. |
|---|---|---|
| A | .157 | 6 |
| B | .101 | 22 |
| C | .062 | 16 |
| D | .052 | 12 |

CARD SIZE, $2.85^{\pm.020}$ X $5.130^{+.002}_{-.008}$




SKETCH "A"




SKETCH "B"

| REV NO. |
|---|
| 0 |

| SCALE | ETCHED CIRCUIT BOARD DWG. | SPACE UNIT REQUIREMENT | |
|---|---|---|---|
| | | FRONT | BACK |
| FULL | 36B605049AA004 | ——— | ——— |

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

PRINTS TO

| | |
|---|---|
| 5QC(BK) | 5 (S) |
| IT(BW) | 5D(BK) |
| | 5(TS)50 |
| | 5B(2)22 |
| | JA(CD) |
| | 5(QC)B |

MADE BY L. Cable JUNE 20, 1973
ISSUED G.R. Baker 3rd FEB. 1, 1974

APPROVALS

SPEED VARIATOR ERIE, PA. (DIV OR DEPT / LOCATION)

36C764129AA
CONT ON SHEET | SH NO.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC

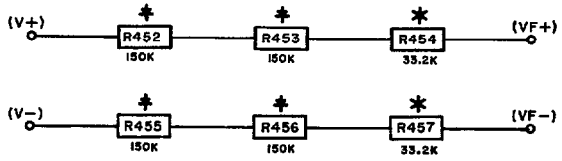
36C764140AA

SCHEMATIC DIAGRAM

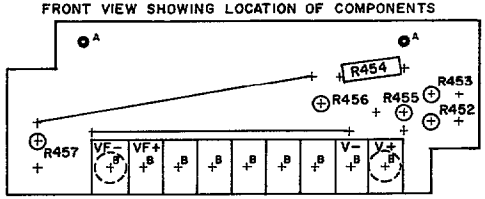
| UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING — | | | | |
|---|---|---|---|---|
| APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS | | |
| | | FRACTIONS | DECIMALS | ANGLES |
| 718A515 | ✓ | ± | ± | ± |

TITLE PRINTED CIRCUIT DIAGRAM
RESISTANCE ISOLATOR CARD

FIRST MADE FOR STANDARD LINE
193X263AAG03




FRONT VIEW SHOWING LOCATION OF COMPONENTS



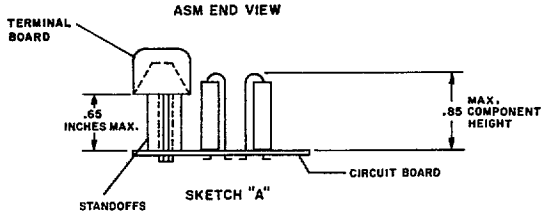

HOLE TABULATION

ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

| LOC | DIA | QUAN |
|---|---|---|
| A | .157 | 2 |
| B | .078 | 9 |

NOTE-1. CARD SIZE, 1.90 $^{+.000}_{-.015}$ X 5.130 $^{+.002}_{-.008}$

ASM END VIEW




SKETCH "A"

NOTES

1. ✱ 1% RESISTOR — 50 PPM
2. ✱ .2% RESISTOR — 50 PPM

| SCALE | ETCHED CIRCUIT BOARD DWG | SPACE UNIT REQUIREMEN | |
|---|---|---|---|
| | | FRONT | BACK |
| FULL | 36B605027AA004 | — | — |

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

PRINTS

| | |
|---|---|
| 5QC(BK)5 | JA(C |
| 5B(2)22 | 5D(B |
| 5(TS)50 | IT(B |
| 5(S) | 5L(B |
| | 5AE(B |
| | 5(QC |

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

MADE BY G R Baker DEC. 17, 1973
ISSUED Paul Novak MARCH 26, 1974

APPROVALS

speed variator
ERIE, PA.

36C764140AA